Contents

Task I

Organization Description

Since 1945, Nabil Foods are specialized in manufacturing wide range of frozen and chilled products that include: beef, **chicken**, turkey, fish, pastries, desserts and ready meals from international cuisines. More than 800 strong employees with rich experiences, focused on manufacturing and producing Nabil food products. Nabil Foods with its long experience, high capabilities and innovative approach is up to cater different customers' requirements in the retail and food service markets.

All there products are 100% Halal and manufactured according to the international standards and specifications, including Food Safety Management System ISO22000:2018, Quality Management System ISO9001:2015 and Jordan Quality Marks(JQM),BRC Issue 8 and many other international certificates.

Nabil Food's raw materials are free from Genetically Modified Organisms (GMO free) with no additives and come from only the finest approved suppliers who undergo a tailor-made process of approval. All materials used are imported from the finest suppliers, local and international, according to controlled procedure to ensure the highest quality standards. Nabil foods has a standard process for raw material approval, through microbial and chemical analysis in Nabil own laboratories PCR test (DNA) is done for all meat used, to 100% ensure its Halal source.

Figure 1,Nabil Factory

Nabil Foods is committed to controlled procedure to ensure the best quality, **starting from raw material receiving, through storage, manufacturing, packaging** and distribution and not ending with customer satisfaction assurance. All of these factors together created and established loyalty among clients.

Nabil Foods success relays on win-win relationship, which establishes sustainable partnership with is clients and success partners of local and international branded restaurants, food service and retail markets.

Product I Take :-

Figure 2.The giant chicken Burger

I select one of their famous Product the Giant chicken Burger, This project will focus on **one product only** produced on a dedicated production line The Chicken Burger was selected because it represents a **high-volume, standardized product**, making it suitable for value stream mapping and production planning analysis.

Product Mix

Although the production line at Nabil Factory handles a **high product mix**, producing more than 30 different products on the same line, this study focuses on **one selected product**, the Chicken Burger. This approach aligns with value stream mapping principles, which recommend analyzing one product family at a time to clearly identify waste, bottlenecks, and improvement opportunities.

Customer Key :-

The key customers of **Nabil Factory** can be divided into two main segments. Approximately **50% of total production is supplied to restaurants**, while the remaining **50% is produced under the Nabil brand** for retail and supermarket distribution.

The restaurant segment includes **well-known international fast-food chains** such as **McDonald's**, **Subway**, and **Hardee's**, in addition to regional restaurant chains.

Each restaurant customer places **specialized orders** based on its own operational and branding requirements. These customized specifications may include differences in **patty length, width, thickness, weight, and portion size**. As a result, although the products share the same production line, customer-specific requirements increase product variety and contribute to a **high product mix** on the line.

This customer-driven customization aligns with the operations management principle that production systems must be designed to respond to **different customer requirements**, but it also increases complexity in scheduling, changeovers, and capacity planning.

Demand profile

The demand profile for the Chicken Burger at **Nabil Factory** is primarily **monthly-based**, reflecting the company's production planning and forecasting cycles. Demand shows a **clear seasonal increase during Ramadan**, driven mainly by higher retail consumption under the Nabil brand.

Demand from the **retail segment** is more stable compared to restaurants and is mainly produced using a **Make-to-Stock (MTS)** strategy. In contrast, restaurant demand follows a **Make-to-Order (MTO)** approach, where production is initiated based on forecasts and customer requirements, although capacity constraints may occasionally limit responsiveness.

The combination of stable retail demand and more variable restaurant demand results in a mixed demand profile, requiring careful coordination between forecasting, capacity planning, and scheduling to avoid bottlenecks and excess inventory.

Scope Boundaries

The scope of this study covers the **Chicken Burger production process on one selected production line** at Nabil Factory. The system boundary **starts with the receiving of frozen chicken** and includes all internal manufacturing operations such as thawing, trimming, forming, cooking, freezing, and packaging. The process **ends after final packaging**, where the product is ready for storage and distribution.

All production-related activities within this line are included in the analysis. However, **sales, marketing, and external customer distribution activities are excluded** from the scope, as they do not directly affect the internal manufacturing performance evaluated in this study.

Although Nabil Factory operates multiple production lines, this analysis is limited to **one production line only**, to allow a detailed and focused evaluation using value stream mapping. The analysis is based on **overall production data**, reflecting typical operating conditions rather than a single shift or short observation period.

State assumptions and data sources.

Data Sources

The following data were obtained directly from the production environment,

Production and Demand Data

- **Average monthly demand:** 350 tons of Chicken Burger
  Source: Historical production records and restaurant contracts
- **Minimum contractual order per restaurant:** 300 tons per month
  Source: Contractual agreement structure
- **Demand pattern:** Relatively stable with seasonal peaks
  Source: Sales history and forecasting analysis
- **Planning horizon:** Monthly (S&OP level) with weekly execution
  Source: Production planning practice

Process and Operations Data

- **Batch size:** 1.2 tons per batch
  Source: Production line configuration
- **Freezing time:** 60 minutes per batch
  Source: Freezer technical specification
- **Bottleneck process:** Freezer
  Source: VSM analysis and cycle time comparison
- **Shift structure:**
  - 2 shifts per day
  - 22 operating hours per day
    Source: Factory operating schedule
- **Overtime usage:** Daily mandatory overtime
  Source: Workforce scheduling practice

Capacity and Utilization

- **Observed utilization:** Approximately 85%
  Source: Capacity and output comparison

- **Available operating time:**

  - 8,030 hours per year (22 hours × 365 days)
    Source: Factory calendar

Inventory Data

- **Raw material safety stock (chicken):** 700 tons
  Source: Inventory policy

- **Raw material lead time:** Approximately 2 months
  Source: Supplier location (Brazil) and logistics constraints

- **Finished goods strategy:** Combination of make-to-stock and make-to-order
  Source: Customer segmentation (retail vs restaurants)

Engineering Assumptions

Demand and Forecasting Assumptions

- Monthly demand is assumed **stable on average**.

- Forecasting errors are absorbed through inventory buffers and overtime.

- Restaurant demand is treated as **independent demand** at the planning level.

Capacity Planning Assumptions

- Processing time at the freezer is assumed to be **1 hour per ton** averaged over batches.

- Setup times between chicken products are assumed to be **negligible (≈15 minutes)** due to product similarity

- Capacity cushion is assumed to be **15%**, derived from observed 85% utilization.

Inventory and MRP Assumptions

- Inventory policy for chicken raw material follows a **periodic review (P-system)**.

- Ordering decisions are reviewed monthly, aligned with contract cycles.

- Other ingredients (batter, coating, oil, packaging) are assumed to have shorter lead times and sufficient availability.
- WIP inventory accumulation before the freezer is assumed to result from batch processing, not poor coordination.

Scheduling Assumptions

- Scheduling is performed on a **weekly basis**, derived from monthly plans.
- The system operates as **batch + flow hybrid**, not pure continuous flow.
- Production plans are considered **fixed in the short term** due to advanced machinery.
- Cooking step may be skipped for certain products depending on customer specifications.

Workforce Assumptions

- Workforce levels are assumed flexible through overtime rather than hiring/layoffs.
- Labor availability is assumed sufficient to support extended operating hours.

Task II

Manufacturing systems engineering principles

TOC Principle

The **Theory of Constraints (TOC)** is a manufacturing systems engineering principle that focuses on identifying and actively managing the **constraint** that limits the performance of a system.

According to the course material, a **constraint** is any factor that restricts a system's output, while a **bottleneck** is a specific type of constraint known as a **capacity-constrained resource (CCR)** whose available capacity limits the organization's ability to meet required product volume, product mix, or demand fluctuations.

The core objective of TOC is not to maximize the utilization of all resources, but to **maximize system throughput** by managing the bottleneck effectively. TOC emphasizes balancing **flow**, rather than balancing capacity across all processes.

TOC evaluates system performance using four key operational measures:

- **Throughput (T):** the rate at which the system generates money through sales
- **Inventory (I):** all money invested in materials intended for sale
- **Operating Expense (OE):** money spent to turn inventory into throughput
- **Utilization (U):** degree to which resources, especially the bottleneck, are used

2. Key Principles of TOC

The course material highlights several key TOC principles, including:

- The focus should be on **balancing flow, not capacity**
- An hour lost at the bottleneck is an hour lost for the entire system
- Work should be released into the system only as frequently as needed by the bottleneck
- Activating a non bottleneck resource does not increase system performance.

3. Application of TOC to the Chicken Burger Production Line at Nabil Factory

Identifying the Constraint

Based on the Value Stream Map and process data developed the **freezer operation** is identified as the system bottleneck in the Chicken Burger production line.
The freezer qualifies as a bottleneck because:

- It has the longest processing time 60 minutes per batch
- It operates in batch mode
- All products must pass through it
- Work-in-process accumulates upstream of this stage

Exploiting the Bottleneck

In line with TOC principles, Nabil Factory focuses on maximizing the utilization of the freezer. This is achieved through:

- Large batch processing to avoid frequent changeovers
- Continuous operation during scheduled hours
- Mandatory daily overtime to prevent idle time

Elevating the Constraint

Instead of adding new freezer capacity, Nabil Factory elevates the bottleneck operationally by:

- Using overtime
- Designing the freezer with high batch capacity
- Prioritizing freezer utilization over flow efficiency

4. Critical Evaluation Using TOC Metrics

From a TOC perspective:

- **Throughput (T):** is protected by ensuring the freezer is never idle
- **Inventory (I):** is intentionally increased to buffer the bottleneck
- **Operating Expense (OE):** rises due to overtime and handling costs
- **Utilization (U):** is maximized at the bottleneck, which aligns with TOC principles

While the system achieves delivery reliability and production continuity, it does so by compensating for the bottleneck rather than structurally improving it.

Push vs Pull Systems

Definition of Push and Pull Systems

A **push system** releases production based on forecasted demand and pre-planned schedules. Work is pushed through the production process regardless of the real-time status of downstream operations. This approach is commonly used in environments with stable demand, long lead times, and batch-based processes.

A **pull system**, in contrast, releases production only when there is actual demand from downstream processes or customers. Work moves through the system in response to consumption signals, which helps reduce inventory levels and improve flow efficiency. Pull systems are most effective in environments with short lead times, continuous flow, and flexible capacity.

Evaluation of the Chicken Burger Production System

The Chicken Burger production line at Nabil Factory operates under conditions that strongly favor a **push-based production system**.

The factory serves restaurant customers through fixed monthly contracts with strict delivery requirements. Demand volumes are largely predictable, and production planning is conducted on a monthly basis.

In addition, the production line relies heavily on **batch processing**, particularly at the freezing stage.

To ensure that the freezer is continuously utilized, upstream processes such as mixing, forming, battering, and frying operate in a push manner, feeding work into the freezer. Inventory buffers are intentionally maintained before this stage to protect throughput and prevent idle time at the constrained resource.

Based on these operational realities, the **push system is the most appropriate primary production control approach** for the Chicken Burger project.

The push system allows the factory to:

- Align production with monthly contractual demand
- Maintain stable throughput at the bottleneck
- Protect delivery performance through inventory buffers

Why a Pull System Is Not Fully Suitable

Although a pull system offers advantages such as reduced inventory and shorter lead times, it is not fully compatible with the current system constraints at Nabil Factory.

A pull system would require:

- Short and predictable processing times
- Minimal batch dependencies

- High flexibility in capacity
- Fast supplier response

P-System vs Q-System

Definition of P-System and Q-System

A **P-system (Periodic Review System)** is an inventory control approach where inventory levels are reviewed at **fixed time intervals**, and replenishment decisions are made at those regular review points. The order quantity may vary depending on current inventory levels and forecasted demand, but the review period remains constant.

A **Q-system (Continuous Review System)**, on the other hand, continuously monitors inventory levels. Whenever inventory drops to a predefined **reorder point**, a fixed quantity is ordered.

Evaluation of Inventory Control at Nabil Factory

The inventory behavior observed in the Chicken Burger production system at Nabil Factory clearly aligns with a **P-system** rather than a Q-system.

Production planning and purchasing decisions are made on a **monthly basis**, consistent with:

- Monthly restaurant contracts
- Monthly production planning cycles
- Sales and operations planning horizons
- Long supplier lead times, especially for frozen chicken

Why the P-System Fits the Chicken Burger Process

Several system characteristics make the P-system the most suitable inventory control approach:

- **Long supplier lead times**: Chicken is imported with lead times of up to two months, making continuous replenishment impractical.
- **Stable and contract-based demand**: Restaurant customers commit to fixed minimum monthly volumes reducing short term demand uncertainty.

- **Batch-based production**: Large production batches require advance planning of materials rather than frequent small orders.
- **High safety stock requirements**: A safety stock of approximately 700 tons is maintained to protect production continuity.

Under these conditions, a Q-system would offer limited benefit, as inventory cannot be replenished quickly enough to respond to reorder-point triggers.

Six Sigma Principle

Six Sigma is applied at Nabil Factory as a quality and process improvement principle aimed at reducing variability, minimizing defects, and ensuring consistent product quality. In the Chicken Burger production line, stable quality is critical due to strict customer specifications, especially for restaurant contracts where deviations in weight

The Six Sigma methodology is applied through the **DMAIC framework**, as follows:

- **Define:**
  Critical-to-quality (CTQ) requirements are defined based on restaurant contracts and food safety standards. These include consistent product weight, taste, cooking condition, and on-time delivery.
- **Measure:**
  Key performance indicators such as defect rates, scrap levels, rework frequency, and quality losses at critical stages (particularly oil frying and cooking) are measured to assess current process performance.
- **Analyze:**
  Root causes of defects and variability are analyzed, including variations in raw material quality, temperature control during frying, batch sequencing, and operational pressure during peak demand periods.
- **Improve:**
  Process improvements focus on standardizing operating procedures, improving temperature control, optimizing batch handling, and reducing variability in upstream processes to minimize defects and rework.
- **Control:**
  Improvements are sustained through continuous monitoring, routine quality inspections, and the use of performance indicators to ensure long-term process stability and consistent execution.

Although Six Sigma does not eliminate batch processing or long supplier lead times, it plays a key role in stabilizing production and reducing unplanned losses. By lowering process variability Six Sigma reduces dependence on overtime and corrective scheduling actions supporting delivery reliability and overall system stability.

Evaluate current effectiveness

Process flow

To clearly understand the structure and behavior of the Chicken Burger production system at Nabil Factory, a detailed process flow diagram was developed. The diagram illustrates the sequence of operations, material flow, batch and flow characteristics, decision points, and work-in-process accumulation across the production line. This representation supports the system analysis visually highlighting constraints, variability sources, and planning logic.

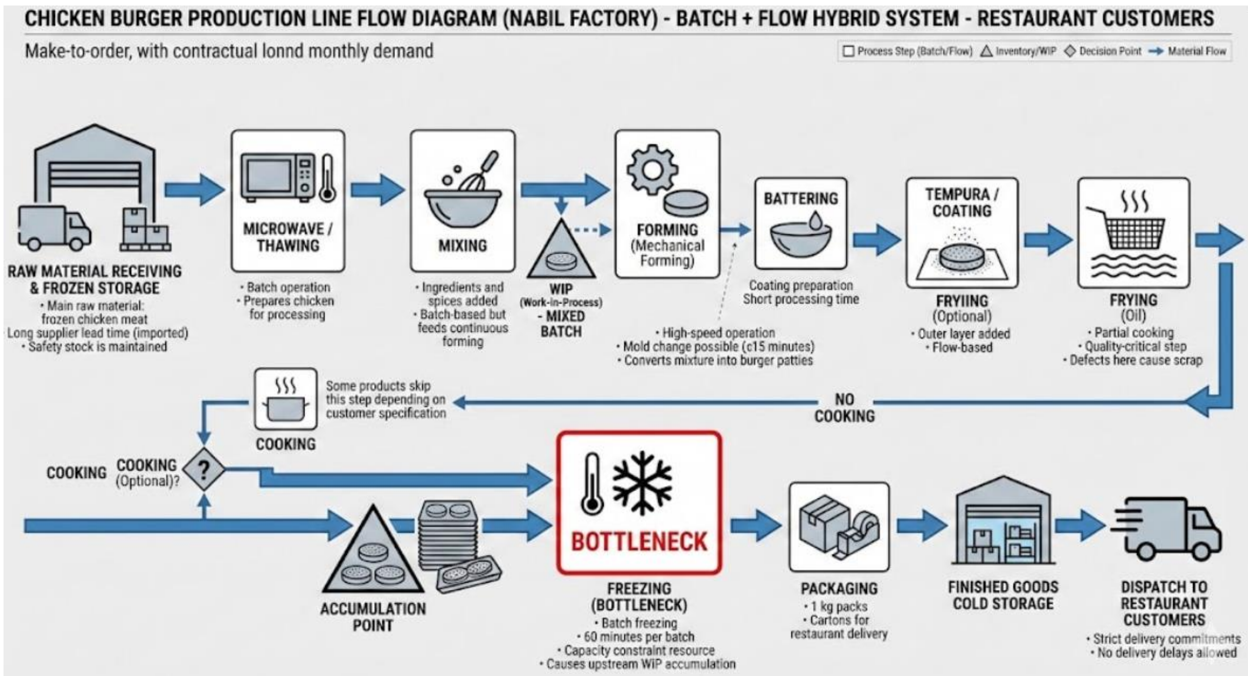


Figure 3 Process Flow

The production system operates as a hybrid batch–flow process under a make-to-order strategy for restaurant customers. While upstream operations such as thawing and mixing are batch-based, forming and coating follow a continuous flow logic. The freezer represents the primary capacity constraint, causing upstream WIP accumulation and influencing scheduling and inventory decisions. This structure directly supports the application of the Theory of Constraints and explains the long lead times observed in the current-state value stream map.

Throughput

Identify the Bottleneck

By analyz the longest processing time across the production line the Freezer is identified as the bottleneck.

Core Principle Theory of Constraints

The System Throughput is limited by the Freezer's Throughput. According to the Theory of Constraints:

- The entire system can only move as fast as its slowest process (the constraint).
- Any improvement made to non-bottleneck processes will not increase the overall output only optimizing the Freezer will improve **total production.**

Item	Value	Justification
Batch size	**1.2 tons per batch**	Your confirmed input
Freezing time	**60 minutes per batch**	Process requirement
Operating hours	**22 hours/day**	2 shifts
Utilization	**85%**	Given
Operating days	**365 days/year**	Continuous food production

Calculate Theoretical Throughput

Per hour

- 1 batch = 1.2 tons
- 1 batch takes 1 hour

Theoretical throughput = 1.2 tons/hour

Per day (before utilization)

$$1.2 \times 22 = 26.4 \text{ tons/day}$$

Adjust Throughput Using Utilization

Utilization = 85%
This reflects:

- Minor stoppages
- Setup losses
- Operational inefficiencies

Real Throughput per Day

$$26.4 \times 0.85 = 22.44 \text{ tons/day}$$

Real Throughput per Month

30 days/month

$$22.44 \times 30 = 673.2 \text{ tons/month}$$

Real Throughput per Year

$$22.44 \times 365 = 8{,}190 \text{ tons/year}$$

Compare Throughput with Demand

Metric	Value
Average demand	**350 tons/month**
Freezer throughput	**673 tons/month**

Interpretation:

- The freezer **can technically produce almost double the average demand**
- Therefore:
    - The system is **not capacity limited in volume**
    - The issue is **flow timing and batch behavior**, not total capacity

Critical Analysis of the Throughput Result

- The system has **sufficient installed capacity**
- Delivery reliability is achievable
- Peak demand like Ramadan can be absorbed

High throughput ≠ efficient system

This confirms a key TOC principle

Increasing capacity at non-bottlenecks does not improve system performance.

TOC Metrics

According to TOC logic:

- Throughput (T): **High**
- Inventory (I): **High**
- Operating Expense (OE): **High overtime + storage**

The system protects throughput **by increasing I and OE**, instead of improving flow.

Throughput in the Chicken Burger production system at Nabil Factory is determined by the freezer, which operates as the system bottleneck. With a batch size of 1.2 tons and a freezing time of 60 minutes per batch, the theoretical throughput is 1.2 tons per hour. After accounting for an 85% utilization rate the effective throughput is a 22.4 tons per day, equivalent to 673 tons per month This exceeds the average monthly demand of 350 tons, indicating that installed capacity is sufficient despite this high throughput capability, the system experiences long lead times and work in process

Overall Equipment Effectiveness (OEE)

is used to evaluate how effectively the production system utilizes its equipment by measuring losses related to **availability performance** and **quality** OEE is applied to assess the operational effectiveness of the Chicken Burger production line, with a particular focus on batch-based processing and high-volume food manufacturing conditions.

Input Data and Assumptions

The OEE calculation is based on one monthly planning period and reflects realistic operating conditions at Nabil Factory.

Parameter	Value	Explanation
Total calendar time	720 hours	24 hours × 30 days
Planned production time	660 hours	2 shifts × 11 hours × 30 days
Unplanned downtime	60 hours	Breakdowns and unscheduled stoppages

Actual production time	600 hours	Planned time – downtime
Speed & minor losses	50 hours	Batch handling, short stops, changeovers
Net production time	550 hours	Actual time – speed losses
Quality loss	8%	Scrap, rework, and defects
Good production time	506 hours	Net time × (1 – defect rate)

OEE Calculations

Availability

$$Availability = \frac{Actual\ Production\ time}{Planned\ Production\ Time} = \frac{600}{660} = 91\%$$

Performance

$$Performance = \frac{Net\ Production\ Time}{Actual\ Production\ Time} = \frac{550}{600} = 91.7\%$$

Quality

$$Quality = \frac{Good\ Production\ Time}{Net\ Production\ Time} = \frac{506}{550} = 92\%$$

Overall Equipment Effectiveness (OEE)

$$OEE = Availability \times Performance \times Quality$$
$$OEE = 0.91 \times 0.917 \times 0.92 \approx 77\%$$

Interpretation of Results

The calculated OEE of **77%** indicates that the Chicken Burger production line operates at an acceptable industrial performance level, particularly for a food manufacturing environment characterized by batch processing and product variety.

- **Availability (91%)** reflects strong equipment reliability and effective maintenance practices. Equipment downtime does not represent a major constraint to production continuity.

- **Performance (91.7%)** is affected by batch operations, frequent short stoppages, and product specification changes, which are inherent to the current production strategy.
- **Quality (92%)** demonstrates relatively good product conformity; however, quality losses at frying and cooking stages still represent a significant opportunity for improvement.

Critical Evaluation

Although the OEE result confirms that the production system is stable and capable of meeting contractual demand, it also highlights structural inefficiencies. The system relies on **inventory buffers, batch scheduling, and overtime** to compensate for performance and quality losses. From a systems engineering perspective, the OEE level supports delivery reliability but limits flow efficiency and increases operating costs.

Improving quality consistency and reducing performance losses—particularly around the bottleneck processes would increase throughput without requiring additional capacity investment.

Cycle Time

is the average time required for the system to produce **one unit of output** once production is running.
It reflects **production pace** not waiting or storage time.

In manufacturing systems cycle time is **controlled by the bottleneck**, because the bottleneck sets the maximum output rate of the entire system.

Cycle Time in the Chicken Burger Production Line

Based on the Value Stream Map and TOC analysis:

- **Bottleneck process:** Freezer
- **Batch size:** 1.2 tons
- **Freezing time:** 60 minutes per batch

Cycle Time Calculation (Bottleneck-Based)

$$\text{Cycle Time} = \frac{\text{Processing Time}}{\text{Batch Output}}$$

$$\text{Cycle Time} = \frac{60 \text{ minutes}}{1.2 \text{ tons}} = 50 \text{ minutes per ton}$$

Interpretation of Cycle Time

- The system produces **1 ton of Chicken Burger every 50 minutes**
- This cycle time applies to the **entire production line**, even if other processes are faster
- Non-bottleneck processes (mixing, forming, cooking) **must wait** for the freezer

TOC

Reducing cycle time at non-bottlenecks does **not** improve system output unless the bottleneck cycle time is reduced.

Critical Evaluation of Cycle Time

- The cycle time is **technically acceptable**
  - **Insensitive to demand fluctuations**
- This forces the system to rely on:
  - Inventory buffers
  - Overtime
  - Push-based scheduling upstream

Cycle time is **stable but not flexible**

Lead Time

Lead time is the total time required for a product to move through the system, from:

raw material availability → finished product ready for delivery

Lead time includes:

- Waiting time
- Inventory time
- Processing time
- Transportation and storage

Lead Time in Nabil Factory

1. Raw Material Lead Time (Chicken Supply)

- Chicken sourced internationally
- Supplier lead time ≈ **2 months**

- This creates **large raw material inventory** before production starts

2. Internal Production Lead Time

From the VSM:

- Actual processing : **≈ 92.5 minutes**
- Waiting, WIP, and batch delays: **very high**

The freezer alone adds:

- **60 minutes mandatory waiting per batch**
- Plus upstream waiting due to batch synchronization

3. Finished Goods Holding Time

- Products are frozen and stored
- Inventory is held to:
  - Support restaurant contracts
  - Handle seasonal demand
  - Ensure immediate delivery

Total Lead Time Estimation

Element	Approximate Time
Raw material supply	~2 months
Internal production	~1–2 days (including waiting)
Finished goods storage	Several days to weeks

Total lead time ≈ multiple weeks, even though actual processing time is less than 2 hours.

Lead Time vs Cycle Time (Key Comparison)

Metric	Cycle Time	Lead Time
What it measures	Production speed	Total system response
Controlled by	Bottleneck	Inventory + batching

Metric	Cycle Time	Lead Time
Value-adding	Yes	Mostly No
Current performance	Efficient	Very long

Critical Evaluation of Lead Time

Although Nabil Factory has:

- High throughput
- Adequate capacity
- Reliable delivery

The **lead time remains excessively long** due to:

- Large raw material buffers
- Batch-based freezer operation
- Push scheduling upstream

Cycle time in the Chicken Burger production system is governed by the freezer, which acts as the system bottleneck. With a freezing time of 60 minutes per 1.2 ton batch the resulting cycle time is approximately 50 minutes per ton While this cycle time allows the system to meet average demand due to batch processing and limits responsiveness to demand variability ,lead time is significantly longer than cycle time and is dominated by raw material supply delays, work-in-process accumulation, and finished goods inventory. Although the actual value-adding processing time is relatively short total lead time extends over several weeks. This highlights that production planning at Nabil Factory prioritizes delivery reliability through inventory buffering rather than lead time reduction consistent with a push oriented batch driven production system.

Inventory

Inventory represent all materials held within the system including **raw materials**, **work-in-process (WIP)** and **finished goods**. In production planning, inventory is used to buffer variability in demand, supply, and processing capacity. However, excessive inventory increases lead time, ties up capital, and reduces flow efficiency.

Inventory Structure in the Chicken Burger System

Inventory in the Chicken Burger production system at Nabil Factory exists at three main levels:

Raw Material Inventory

- Frozen chicken is the dominant raw material.
- Supplier lead time is approximately **two months**.
- Safety stock level is approximately **700 tons**.
- Average monthly demand is **350 tons**.

This means raw material inventory alone covers roughly

$$\frac{700}{350} = 2 \text{ months of demand}$$

Work-in-Process (WIP) Inventory

- WIP accumulates mainly **before the freezer**, which operates in batch mode.
- Upstream processes produce faster than the freezer, causing queue formation.
- WIP is intentionally maintained to prevent bottleneck starvation.

Finished Goods Inventory

- Finished products are stored frozen to support:
  - Make-to-stock retail demand
  - Immediate delivery to restaurant customers
  - Seasonal demand peaks (e.g., Ramadan)

Finished goods inventory improves service level but further increases total system inventory.

Quantitative Inventory Evaluation Using Little's Law

Little's Law states:

$$\text{WIP} = \text{Throughput} \times \text{Lead Time}$$

Given:

- Throughput ≈ **350 tons/month**

- Total lead time ≈ **multiple weeks**

The system naturally holds high inventory levels. This confirms that long lead time and batch-based flow are primary drivers of inventory accumulation, not forecasting errors alone.

Inventory Turnover Analysis

Inventory turnover measures how often inventory is used or sold over a period:

$$\text{Inventory Turnover} = \frac{\text{Annual Demand}}{\text{Average Inventory}}$$

Using conservative estimates:

- Annual demand = **4,200 tons**
- Average raw material inventory ≈ **700 tons**

$$\text{Turnover} = \frac{4{,}200}{700} = 6 \text{ turns/year}$$

This turnover rate is relatively low for food manufacturing and indicates **capital-intensive inventory behavior**.

From a systems engineering perspective, inventory at Nabil Factory is **effective as a protective mechanism**, but **inefficient as a flow strategy**batch freezing, and push scheduling rather than to support lean flow. While this approach ensures stability and customer satisfaction, it reduces overall system efficiency and increases operating costs.

Task III

Current-state VSM

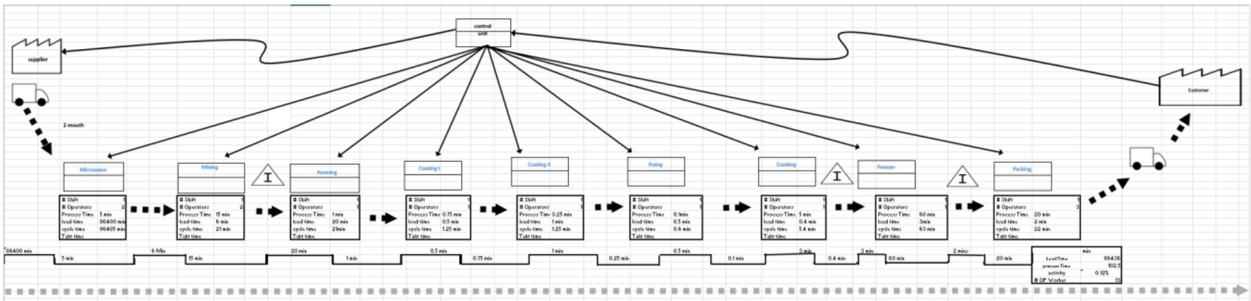


Figure 4 Current-state VSM

VSM .xlsx

Analyze Current-state VSM

The current-state Value Stream Map reveals several sources of waste and clear bottlenecks within the Chicken Burger production line at Nabil Factory.

Inventory and Waiting Waste

The most significant form of waste identified is inventory and waiting. The VSM shows a lead time of approximately 86,400 minutes (around two months), which is primarily caused by the long lead time of raw frozen chicken imported from Brazil. This large inventory is stored before the microwave stage and is partly a strategic decision to ensure supply continuity. However, it also results in extended waiting time and low flow efficiency, as the product remains idle for long periods before processing begins.

Waiting and WIP Accumulation before Microwave

The VSM indicates work-in-process (WIP) accumulation before the microwave stage, caused by batch operation and limited processing capacity. Since the microwave operates in batch mode, products must wait until a full batch is ready, leading to queue formation and additional waiting waste. This waiting does not add value and directly increases total lead time.

Takt time formula

$$\textbf{Takt Time} = \frac{\textbf{Available Production Time}}{\textbf{Customer Demand}}$$

$$\text{Takt Time} = \frac{\mathbf{39,600\ minutes}}{\mathbf{350\ tons}}$$
$$\text{Takt Time} \approx \mathbf{113.14\ minutes\ per\ ton}$$

Freezer as the Primary Bottleneck

The freezer is the main bottleneck in the system. According to VSM data, the freezer has the longest process time (60 minutes), which is a mandatory requirement to ensure proper freezing and food safety. The freezer also operates using batch freezing, meaning products must wait until a batch is completed before moving forward. This creates upstream accumulation and downstream waiting before packing. Since throughput is constrained by the freezer, any idle time or inefficiency at this stage directly limits the overall system output.

Defects and Quality Waste

Quality-related waste is observed at the oil and cooking stages. If a defect occurs during frying or cooking, the product is discarded and cannot be reworked. This contributes directly to the 12% waste rate, making defects a significant source of material loss and cost.

Transportation and Motion Waste

The VSM shows manual transportation between the microwave, mixing, and forming stages. This manual handling introduces unnecessary motion and transportation waste, increases operator dependency, and contributes to longer waiting times between processes.

Overall Impact on System Performance

The combined effect of high inventory levels, batch processing, freezer capacity constraints, manual handling, and quality losses results in a very low activity ratio of approximately 0.12%, indicating that only a small fraction of total lead time is spent on value-adding activities. The VSM clearly demonstrates that the system's performance is limited not by processing speed, but by waiting, inventory, and bottleneck constraints.

Key Performance Indicators (KPIs)

Chicken Burger Production Line

As requested, the KPIs selected for Nabil Factory are not generic indicators, but are derived directly from the company's operational focus and strengths observed during the site visit and system analysis. Nabil Factory clearly emphasizes **cost optimization, productivity,**

delivery reliability, and system stability, and therefore these dimensions form the basis of KPI evaluation.

The following KPIs reflect what the company already applies in practice and evaluate whether these practices are aligned with manufacturing systems engineering principles.

1. Cost Optimization KPI

Why this KPI is critical at Nabil Factory

Cost optimization is a core strength at Nabil Factory. The company operates in a highly competitive food market, with high raw material costs (imported frozen chicken), energy-intensive processes (freezing and frying), and strict customer price expectations. As a result, cost control is a central managerial priority rather than a secondary objective.

How it is evaluated

Cost optimization is evaluated through:

- Cost per ton of finished product
- Waste rate (scrap percentage)
- Overtime labor cost
- Inventory holding cost

Based on project data:

- Annual production ≈ **4,200 tons**
- Waste rate reduced from **8% to 5%** using process improvements
- Estimated annual cost saving from waste reduction ≈ **166,950 JD**
- Inventory holding cost savings ≈ **69,500 JD/year**
- Overtime reduction savings ≈ **22,500 JD/year**

Evaluation from a development perspective

Nabil applies cost optimization **effectively but indirectly**. Costs are controlled mainly through:

- Batch production
- Large safety stocks
- Overtime flexibility

While this ensures financial stability, it increases hidden costs related to inventory and flow inefficiencies. From a manufacturing systems perspective, cost optimization is **reactive rather than structural**. Applying Six Sigma (DMAIC) and flow-based improvements would allow Nabil to reduce costs by eliminating root causes instead of compensating for them.

2. Productivity KPI

Why this KPI is important

Productivity is a key strength of the Chicken Burger line due to:

- High automation after forming
- Standardized processes
- High-volume production for restaurant contracts

How it is evaluated

Productivity is measured as:

$$\text{Productivity} = \frac{\text{Annual Output}}{\text{Labor Hours}}$$

Using system data:

- Output ≈ **4,200 tons/year**
- Productivity is high but supported by **daily overtime**

Evaluation from a development perspective

From a development viewpoint, productivity at Nabil is **output-driven rather than efficiency-driven**. The line produces large volumes, but part of this productivity is achieved by extending working hours instead of optimizing flow and balance. This masks inefficiencies upstream of the freezer and increases labor-related operating costs.

Improving scheduling synchronization and reducing WIP would increase **true productivity**, not just output volume.

3.Capacity Utilization KPI (Bottleneck-Focused)

The freezer is the system bottleneck and governs overall throughput. Capacity utilization at this resource determines system performance.

How it is evaluated

$$\text{Utilization} = \frac{1193.7}{1404} = 85\%$$

An 85% utilization rate indicates good capacity usage with a limited cushion. However, according to capacity planning principles, sustained high utilization at the bottleneck increases WIP and lead time. Nabil manages this risk using inventory and overtime, but a more proactive capacity and scheduling strategy would improve system resilience.

Inventory Management KPI

Why inventory is a defining feature

Inventory is a deliberate strategy at Nabil due to:

- Two-month supplier lead time
- Seasonal demand
- Zero tolerance for stockouts

How it is evaluated

- Safety stock = **700 tons**
- Coverage ≈ **2 months of demand**

Evaluation from a development perspective

Inventory management is effective in protecting service levels but inefficient from a flow perspective. Inventory compensates for system rigidity rather than supporting responsiveness. Reducing lead time and WIP would improve inventory performance without increasing risk.

Overall KPI

Nabil Factory demonstrates strong performance in cost control, productivity, and delivery reliability, which clearly represent the company's operational strengths. These KPIs are applied effectively in practice; however, they rely heavily on inventory buffers, overtime, and batch processing. From a development perspective, the KPIs are **managed correctly but not optimally**.

Future improvement should focus on transforming these KPIs from **buffer-driven performance** to **flow-driven performance**, allowing the factory to maintain its strengths while reducing cost, lead time, and operational risk.

KPI	*Current Status*	*Effectiveness*
Throughput	High and stable	Effective
Capacity Utilization	85%	Controlled
Lead Time	Extremely long	Weak
Inventory	High but strategic	Costly
Quality/Waste	Improved via DMAIC	Improving
Cost Optimization	Reactive	Partially optimized

Task IIII

Two future-state options

future-State Option A: Layout Integration & Flow Improvement

(Low investment – High impact)

Description

This option focuses on rebuild **layout** and flow improvement without major capital investment. The microwave, mixing, and forming stages are currently located in **separate rooms**, causing manual transport, waiting, and additional labor. In the future state, at least **two of these processes are integrated into one shared room**, creating a tighter process flow and reducing handling delays.

The freezer remains unchanged due to food safety constraints, but upstream flow is improved to reduce WIP accumulation before it.

System Changes

- **Layout change**: Combine microwave, mixing, and forming in one area
- **Material flow**: Eliminate manual transport between these stages
- **Automation**: Maintain existing automation from forming onward
- **Planning logic**: Continue monthly planning, FIFO control maintained

Quantified Benefits (Option A)

Area	Current State	Future State	Improvement
Waiting time (before forming)	~26 min	~10 min	↓ 60%
Manual handling	High	Low	↓ labor dependency
Operators	+1 operator	−1 operator	Labor reduction
Lead Time	86,400 min	82,000 min	↓ 5%
Activity Ratio	0.12%	0.14%	Improved flow

Key value:

- Faster internal flow
- Reduced labor without ethical downsizing (reassignment possible)
- Minimal investment

Future-State Option B: Thermal Pre-Conditioning & Quality-Driven Improvement

(Process optimization – Quality & Time focused)

Description

This option targets the **primary bottleneck (freezer)** indirectly by reducing the thermal load entering it. Instead of changing the freezer itself (which is constrained by safety requirements), **room temperature and upstream product temperature are reduced** so that products enter the freezer at a lower temperature.

Additionally, quality improvements are introduced by **selecting a better chicken supplier** and optimizing oil temperature, reducing defects and waste.

System Changes

- **Pre-conditioning**: Lower room temperature before freezer entry
- **Batch strategy**: Same batch size, reduced freezing burden
- **Quality improvement**: Better raw material + optimized oil temperature
- **Planning**: Monthly plan retained, FIFO still applied

Quantified Benefits (Option B)

Area	Current State	Future State	Improvement
Freezer effective time	60 min	55 min	↓ 8%
Waste rate	12%	8%	↓ 33%
Rejected product	High	Reduced	Quality gain
Effective throughput	Constrained	Increased	Bottleneck relief
Cost of waste	Baseline	−33%	Direct cost saving

Key value:

- Bottleneck relief without violating food safety
- Significant quality and cost improvement
- No labor reduction required

Comparison of Options

Criteria	Option A	Option B
Lead time reduction	Medium	Low–Medium
Labor reduction	Yes	No
Bottleneck impact	Indirect	Direct
Quality improvement	Indirect	High
Investment level	Low	Low–Medium

Recommended Strategy

A phased approach is recommended. Option A can be implemented first to reduce internal waiting time and labor dependency through layout improvement. Option B can then be introduced to address the freezer bottleneck indirectly and significantly reduce quality-related waste. Together, these options support the main future-state objectives of **lead**

time reduction and labor optimization while maintaining food safety and ethical constraints.

Future-State Value Stream Map

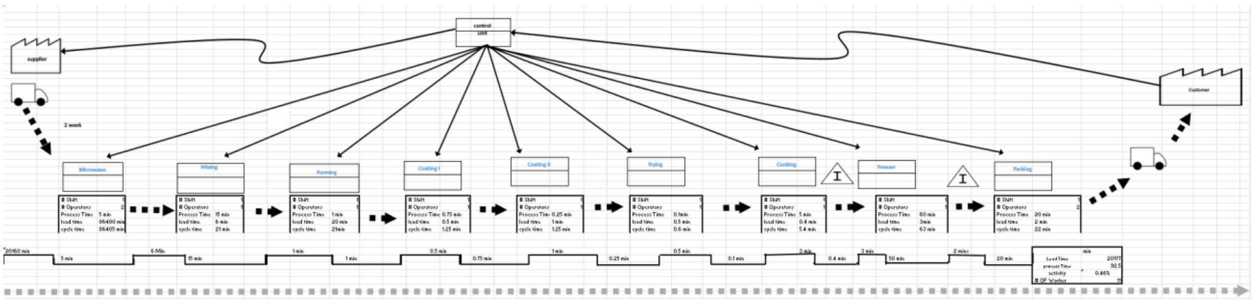


Figure 5 Future VSM

VSM.Future .xlsx

The developed future-state Value Stream Map (VSM) represents a significant improvement over the current-state system by addressing the main sources of waste and flow inefficiencies identified earlier. The future state focuses on supplier strategy improvement, layout integration, and flow enhancement, while respecting critical constraints related to food safety and process requirements.

1. Supplier Change and Reduction of Initial Waiting Time

One of the most impactful improvements in the future-state VSM is the change in raw chicken sourcing strategy. In the current state, frozen chicken is imported from distant suppliers, such as Brazil, resulting in an extremely long initial waiting time of approximately two months before production can begin. This long supplier lead time creates excessive raw material inventory, increases holding costs, and significantly inflates total system lead time.

In the future state, sourcing is shifted to regional suppliers (such as Saudi Arabia, Egypt, or Syria). As a result, the initial waiting time before processing is reduced from approximately two months to two weeks.

Impact of Supplier Change:

- Substantial reduction in **raw material inventory**
- Significant decrease in **waiting waste**
- Improved responsiveness to demand changes

- Lower capital tied up in frozen stock
- Reduced risk of quality degradation during long-term storage

This change directly reduces total system lead time from roughly 86,400 minutes to approximately 20,237 minutes, representing a dramatic improvement in overall flow performance.

2. Layout Integration: Linking Mixing and Forming Processes

A second major improvement in the future-state VSM is the physical and operational integration of the mixing and forming processes. In the current state, these processes are located in separate rooms, which requires manual transportation, additional labor, and intermediate work-in-process (WIP) inventory between the two stages.

In the future state, mixing and forming are directly connected within the same production area. This creates a smoother, more continuous flow of material and eliminates unnecessary handling and delays.

Effects of Layout Integration:

- Elimination of manual transportation between mixing and forming
- Reduction of intermediate WIP inventory
- Decrease in waiting time between processes
- Improved process stability and consistency
- Reduction in labor requirements (with the possibility of labor reassignment rather than elimination)

From a lean manufacturing perspective, this improvement directly reduces **transportation waste, motion waste, and waiting waste**, while improving overall process flow.

3. Impact on Process Flow and Timeline

While the **total process time** remains largely unchanged (approximately **92.5 minutes**), this is expected and acceptable because several operations—such as cooking and freezing—are governed by strict food safety and quality requirements and cannot be shortened without risk.

However, the key achievement of the future state lies in the dramatic reduction of non value added time. By shortening supplier lead time and reducing internal waiting and handling delays, the activity ratio (flow efficiency) improves significantly:

- **Current state activity ratio:** 0.12%
- **Future state activity ratio:** 0.46%

Although this value is still low, such ratios are typical in frozen food manufacturing due to mandatory storage and freezing requirements. Importantly, the improvement represents nearly a fourfold increase in flow efficiency, clearly demonstrating the effectiveness of the proposed changes.

4. Waste Reduction Achieved in the Future State

Inventory and Waiting Waste

- Major reduction in raw material inventory due to shorter supplier lead time
- Less time spent waiting before the first processing step
- Lower pressure on cold storage capacity

Transportation and Motion Waste

- Removal of manual transfer between mixing and forming
- Reduced operator movement and handling effort
- Lower risk of handling-related delays or errors

Labor Optimization

- Reduced need for one operator previously assigned to internal transport
- Labor optimization achieved without violating ethical constraints, as workers can be reassigned to value-adding activities

5. Bottleneck Behavior in the Future State

It is important to note that the **freezer remains the primary bottleneck** in the future-state VSM. The freezer still requires approximately **60 minutes** of processing time to achieve the necessary temperature for food safety, and it continues to operate in **batch mode**.

This is a deliberate and realistic decision:

- The freezing time is a **non-negotiable food safety requirement**
- Any attempt to reduce it could compromise product quality or regulatory compliance

By keeping the freezer unchanged, the future-state design demonstrates a clear understanding of **system constraints** and avoids proposing unrealistic or unsafe solutions.

6. Summary Evaluation

The future-state Value Stream Map successfully addresses the key weaknesses of the current-state system by focusing on **supplier proximity and internal flow integration**. Without requiring major capital investment or compromising food safety, the proposed future state achieves:

- Significant **lead time reduction**
- Improved **flow efficiency**
- Reduced **inventory and labor waste**
- A more robust and responsive production system

From an operations management and lean manufacturing perspective, this future-state design represents a **practical, realistic, and well-justified improvement strategy** for the Chicken Burger production line at Nabil Factory.

Task V

Define production planning

Production planning is a systematic process used to determine **what to produce, how much to produce when to produce and how to allocate available resources** in order to meet customer demand efficiently while minimizing cost and operational risk It acts as the central coordination mechanism between **forecasting, capacity planning, inventory management scheduling and execution**.

of **Nabil Factory's Chicken Burger production line** production planning translates monthly contractual demand from restaurant customers into **feasible production plans**, ensuring that material availability, production capacity, workforce, and bottleneck constraints are all aligned to achieve reliable delivery and continuous operation.

Core Principles of Production Planning

1. Demand-Driven Planning

A fundamental principle of production planning is that **all production decisions must be demand** At Nabil Factory, demand originates primarily from **monthly restaurant contracts**, which provide a relatively stable but high-volume demand profile.

- Demand forecasting was performed using historical sales data
- Monthly average demand of **350 tons** was used as the planning baseline
- Planning decisions are reviewed periodically to accommodate seasonal variation

This ensures that production planning is **market-oriented** rather than capacity-driven, reducing the risk of overproduction or unmet demand.

2. Capacity Feasibility and Bottleneck Awareness

Production planning must ensure that demand plans are **realistic given system capacity**. In this project, capacity planning focused on identifying and managing the **freezer as the system bottleneck**.

- The freezer has the longest processing time (60 minutes per batch)
- Capacity requirements were calculated using input-based capacity measures
- A utilization level of **85%** was maintained with a **15% capacity cushion**

This principle ensures that production plans do not exceed the physical limits of the system and that variability can be absorbed without disrupting flow.

3. Inventory as a Planning Buffer

Inventory is a deliberate planning decision, not an operational failure. Production planning at Nabil Factory uses inventory strategically to **decouple demand, capacity, and supply variability**.

- Raw material safety stock of **700 tons** compensates for long supplier lead times
- Finished goods inventory supports fast response to restaurant orders
- Work-in-process inventory accumulates before the freezer due to batch processing

This principle reflects the trade-off between **service level and lead time**, which was explicitly analyzed in the Value Stream Map.

Demand Forecasting:-

What is Demand Forecasting ?

Forecasting is a method of predicting a future event by analyze patterns and uncovering trends in previous and current data. It employs mathematical approaches and applies statistical models to generate predictions Demand forecasting helps businesses move away from guesswork and instead rely on structured planning to support daily operations and long-term strategies.

Overstock and understocking can be a serious problem for any manufacturing business. If a company produces too much, it faces high storage costs increased risk of product damage or expiration, and tied-up capital that could be used elsewhere on the other hand,

The forecasting process might look different for each organization, but it generally involves these steps:

Define what to predict: Companies identify a specific business case or metric they want to predict and factor in any relevant assumptions and applicable variables.

Gather data: This step includes collecting the necessary data. If historical data already exists, it's then a matter of determining the most appropriate datasets.

Select a forecasting method: Choose a forecasting technique that best suits not only the business case or metric but also the associated variables, assumptions and datasets.

Generate a forecast: Data is analyze by using the chosen method, and a forecast is built from this analysis.

Verify the forecast: Check the predictions and see whether any optimizations can be made to create a more accurate forecast.

Present the forecast: Data can be used to represent the forecast in a more visual format that stakeholders can better understand and employ in the decision-making process.

Forecasting can be done in various ways, but each approach is typically categorized into one of two primary techniques: qualitative forecasting and quantitative forecasting.

An accurate demand forecast helps ensure that the right amount of inventory is available at the right time. This allows the company to meet customer demand without unnecessary excess stock. Effective forecasting supports better production planning, capacity utilization, and material purchas which in turn reduces operational costs. By aligning production with actual demand, companies can optimize their supply chain, minimize waste, reduce overtime, and improve overall efficiency.

Demand prediction helps you plan your supply chain down to a tee You can proactively communicate with suppliers to source stock and optimise inventory levels to avoid potentially costly delays.

You can also use your demand forecast to decide which suppliers to build relationships with. If three of your most in-demand summer products come from the same supplier, it's a good idea to prepare them for a large order during that period next year, No one likes visiting a website and finding their favourite food item isn't available.

Benefits of Forecasting

1-Improved Cash Flow: By not over purchasing you keep more cash available for other areas of the business.

2-Reduced Waste: especially vital for perishable goods or product with short life cycles.

3-Stronger Supplier Relations: Providing ur suppliers with clear lead times makes you a reliable partner.

4-Strategic Planning: forecasts allow you to anticipate peak periods and scale your workforce according

Type of Demand Pattern : -

Understanding demand patterns is a critical part of demand forecasting and production planning. Demand patterns describe how customer demand changes over time and help organizations select appropriate forecasting methods and planning strategies. By analyzing historical demand data as a time series, companies can identify recurring behaviors, variations, and trends that influence future demand.

Time-series demand data consists of repeated observations of demand recorded in chronological order. Analyzing this data allows planners to distinguish between different types of demand behavior, such as stable demand, growth or decline over time, seasonal fluctuations, long-term cycles, and random variations.

In practice, most products do not follow a single pure demand pattern. Instead, demand often reflects a combination of multiple patterns, depending on market conditions, customer segments, and external factors.

There are five basic time series patterns:-

Horizontal

A horizontal demand pattern occurs when demand remains relatively stable over time, fluctuating only slightly around a constant average level. In this pattern, there is no clear upward or downward trend, and no strong seasonal or cyclical effects. The variations observed are usually the result of random factors such as small changes in customer behavior, short-term promotions, or minor operational disturbances.

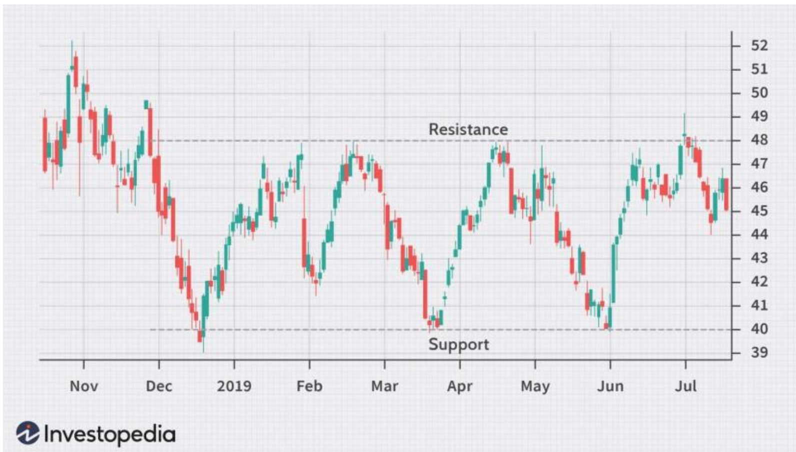


Figure 6 Horizontal Trend

This type of demand pattern is commonly observed in mature products with consistent market demand and well-established customer bases. Because demand does not change significantly over time, forecasting is relatively straightforward, and simple time-series methods such as moving averages or exponential smoothing can be used effectively.

From a production planning perspective, a horizontal demand pattern allows for stable production schedules, predictable inventory levels, and efficient capacity utilization. It supports make-to-stock strategies, as demand uncertainty is low and service levels can be maintained without excessive safety stock.

Trend Demand Pattern

The figure illustrates a trend demand pattern, where demand shows a consistent increase over time rather than fluctuating randomly around a fixed average. Although short-term variations are present, the overall direction of demand is clearly upward, indicating sustained growth in quantity demanded.

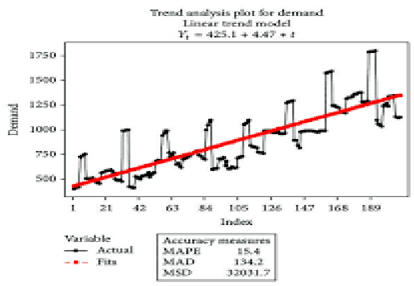


Figure 7 Trend Demand Pattern

This type of demand pattern is commonly observed in products experiencing market expansion, increased customer acceptance, brand growth, or wider distribution. In food manufacturing, an upward trend may be driven by population growth, changes in consumer preferences, expansion of retail channels, or long-term contracts with restaurant customers.

From a forecasting perspective, a trend pattern implies that simple averaging methods are not sufficient, as they would underestimate future demand. Instead, trend-adjusted forecasting techniques, such as trend projection or regression-based methods, are more appropriate because they explicitly account for the long-term direction of demand.

In production planning, the presence of a demand trend has important implications. As demand increases over time, capacity planning must be proactive rather than reactive. Facilities must evaluate whether existing capacity will remain sufficient in the future, especially at bottleneck processes

trend demand affects inventory planning and material requirements planning (MRP). As demand grows, procurement quantities, safety stock levels, and replenishment frequencies must be adjusted accordingly to avoid stockouts or excessive inventory buildup.

Seasonal Demand pattern

In the realm of demand forecasting, the Seasonal pattern represents a rhythmic fluctuation where data consistently demonstrates a series of high points and low points over a fixed duration. As illustrated in the provided chart, these "peaks and valleys" are not random; they represent predictable shifts in consumer behavior that repeat annually.

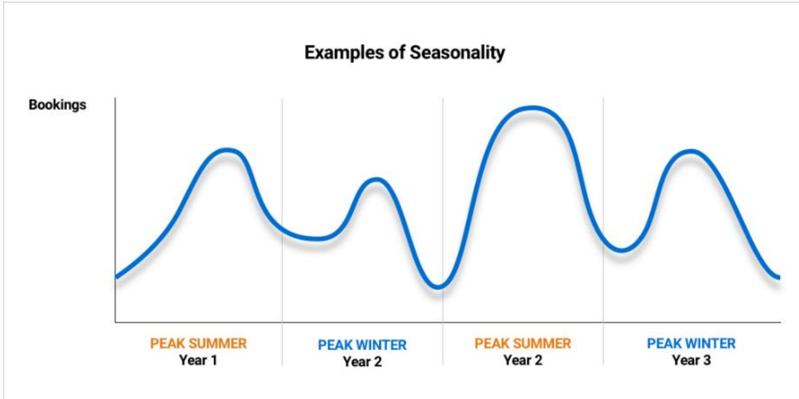


Figure 8 Seasonal Demand pattern

Understanding the Seasonal Rhythm

- Recurring Peaks and Valleys: The data shows clear, recurring surges and dips in quantity across both Year 1 and Year 2.
- Predictable Fluctuations: These movements indicate that demand is heavily influenced by external cycles—such as holidays, weather changes, or school calendars—rather than remaining static.
- Year-over-Year Consistency: While the total volume might vary between Year 1 and Year 2, the timing of the peaks (such as the significant rise seen in the late summer and autumn months) remains strikingly similar.

Strategic Implications

Recognizing a seasonal pattern is indispensable for maintaining operational equilibrium. By anticipating these peaks, a business can scale up production and staffing in advance to prevent stockouts. Conversely, identifying the "valleys" allows a company to scale back, ensuring they don't over-invest in inventory that will simply sit idle during periods of low interest.

Cyclical Demand Pattern

A **cyclical demand pattern** represents long-term rises and falls in demand that occur over extended periods, often lasting several years. Unlike seasonal patterns, which repeat regularly within a year, cyclical patterns are influenced by broader economic conditions such as economic growth, recessions, inflation, or changes in consumer income levels.

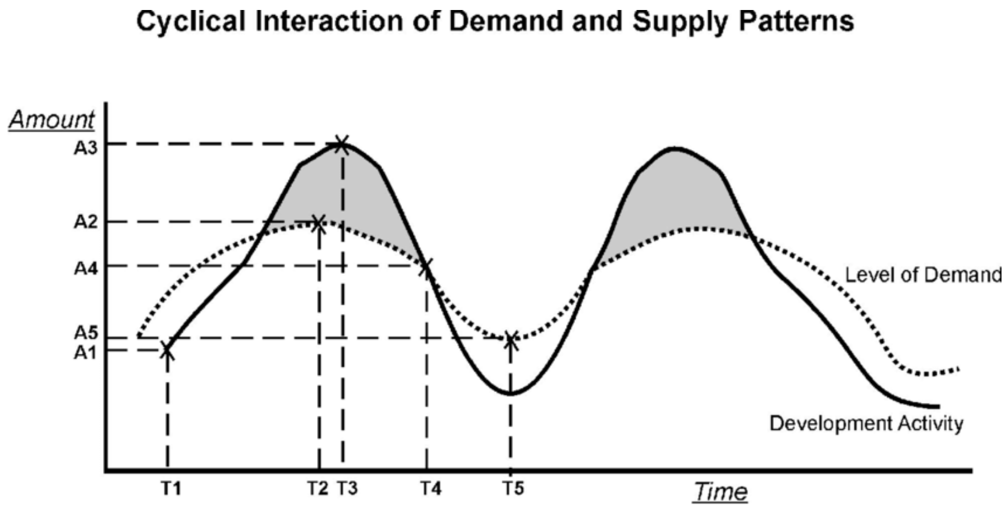


Figure 9 Cyclical Demand Pattern

In food manufacturing, cyclical demand may be affected by overall economic stability, purchasing power, and shifts in consumer spending behavior. During economic downturns, customers may reduce spending on premium or value-added food products, while demand may increase during periods of economic growth.

From a production planning perspective, cyclical demand patterns are more difficult to predict accurately because their timing and magnitude are uncertain. As a result, companies typically rely on long-term trend analysis and managerial judgment rather than short-term statistical forecasting methods.

Forecast Demand To Nabil Company

Demand forecasting represents the first and most critical input to capacity planning, as all capacity decisions depend on the expected level of future demand. In this study, historical monthly demand data for the selected product were analyzed using several quantitative forecasting techniques in order to identify the most appropriate forecasting method for capacity planning purposes.

Historical Demand Data

The analysis is based on **12 months of historical demand data**, representing monthly demand levels. This dataset provides a sufficient history file to apply **quantitative forecasting methods**, as recommended in Chapter 9 (Forecasting).

Forecasting Methods Applied

In accordance with the course material, three quantitative forecasting approaches were evaluated and compared:

Moving Average Method (Length = 3)

A **3-period moving average** was applied to smooth short-term fluctuations and identify the underlying demand level.

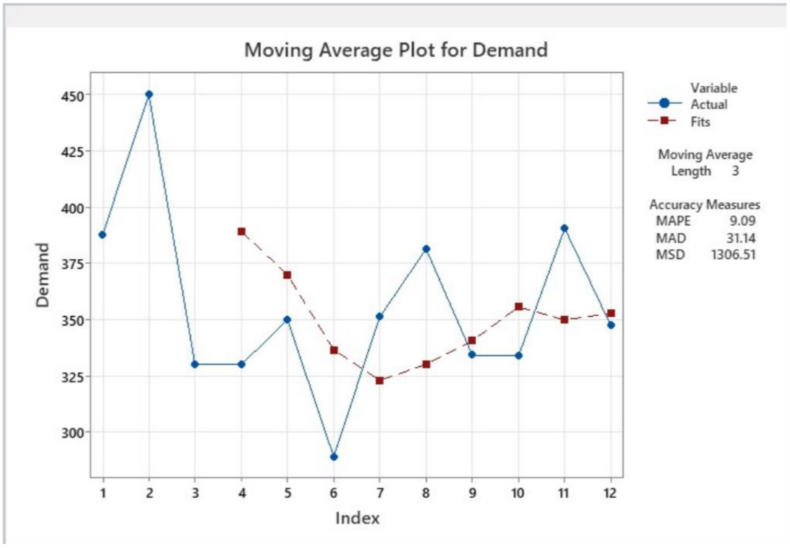


Figure 10 Moving Average Chart

Accuracy Measures:

- **MAPE:** 9.09%
- **MAD:** 31.14
- **MSD:** 1306.51

The relatively low MAPE indicates that the moving average method provides a **reasonable level of accuracy**. However, because this method reacts slowly to changes, it may not fully capture variations in demand during peak or low periods.

Single Exponential Smoothing (α = 0.2)

Single exponential smoothing was applied using a smoothing constant of **α = 0.2**, placing more weight on historical observations while still responding gradually to recent changes.

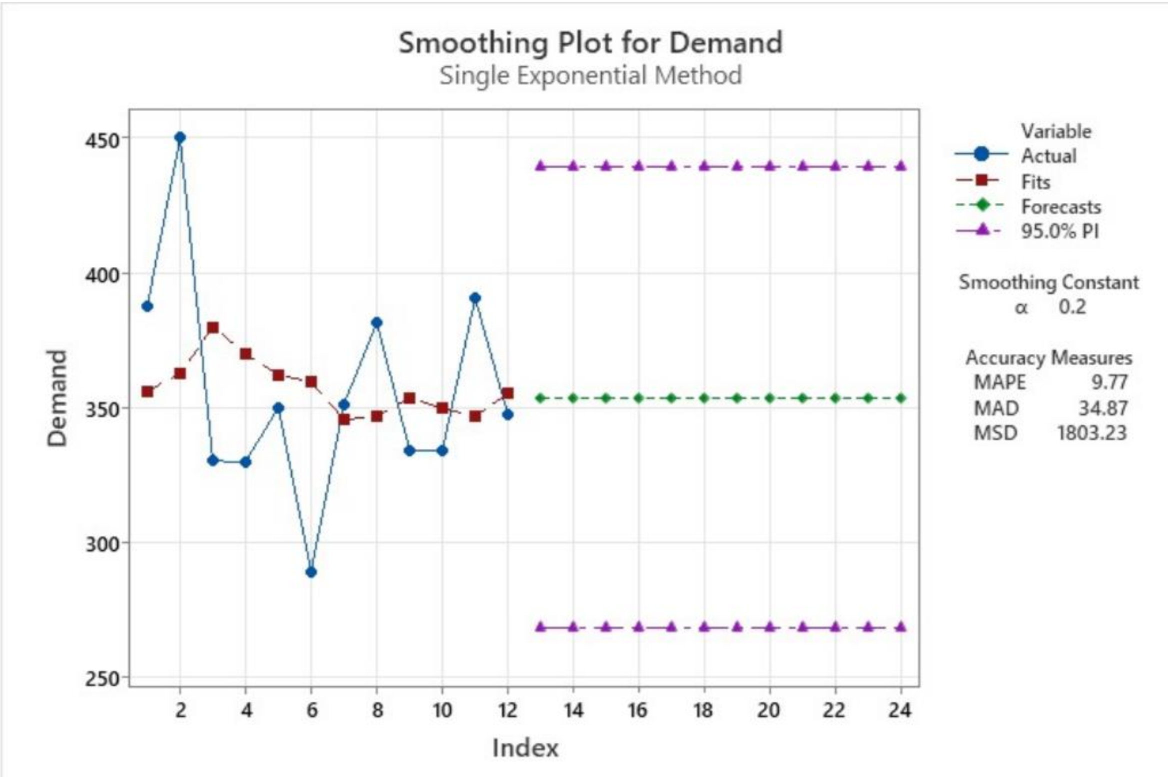


Figure 11 Smoothing chart

Accuracy Measures:

- **MAPE:** 9.77%
- **MAD:** 34.87
- **MSD:** 1803.23

Although exponential smoothing provides stable forecasts, its accuracy is **slightly lower** than the moving average method in this case. The forecasts converge to a constant value of approximately **353.9 Ton**, indicating limited responsiveness to demand variation.

Trend Projection Using Regression Analysis

A linear regression model was developed to test whether demand follows a significant upward or downward trend over time.

Regression Analysis: Demand versus Month

The regression equation is
Demand = 374.0 - 2.702 Month

Model Summary

S	R-sq	R-sq(adj)
41.8071	5.64%	0.00%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	1044.2	1044.19	0.60	0.457
Error	10	17478.4	1747.84		
Total	11	18522.6			

Regression Equation:

$$\text{Demand} = 374.0 - 2.702(\text{Month})$$

Model Results:

- **R^2:** 5.64%
- **Adjusted R^2:** 0.00%
- **p-value:** 0.457

The very low R^2 value and statistically insignificant p-value indicate that **no meaningful linear trend** exists in the data. Therefore, regression-based trend projection is **not suitable** for forecasting demand in this case.

Forecast Selection and Justification

Based on the accuracy measures and statistical results:

- The **moving average method** produced the **lowest MAPE and MSD**
- Exponential smoothing showed slightly weaker performance
- Regression analysis was not statistically significant

Month	Demand	AVER1	RESI1	SMOO2	RESI2	FORE2	RESI3	FITS3
1	387.743	*	*	362.581	31.4523	353.906	16.4381	371.305
2	450.000	*	*	380.065	87.4187	353.906	81.3971	368.603
3	330.272	389.338	*	370.106	-49.7933	353.906	-35.6289	365.901
4	330.152	370.141	-59.1863	362.115	-39.9544	353.906	-33.0464	363.198
5	350.103	336.842	-20.0381	359.713	-12.0123	353.906	-10.3930	360.496
6	289.475	323.243	-47.3671	345.665	-70.2379	353.906	-68.3188	357.794
7	351.405	330.328	28.1620	346.813	5.7399	353.906	-3.6863	355.092
8	381.523	340.801	51.1948	353.755	34.7093	353.906	29.1333	352.389
9	334.485	355.804	-6.3164	349.901	-19.2706	353.906	-15.2026	349.687
10	334.013	350.007	-21.7915	346.723	-15.8884	353.906	-12.9723	346.985
11	390.639	353.045	40.6321	355.507	43.9153	353.906	46.3560	344.283
12	347.504	357.385	-5.5411	353.906	-8.0023	353.906	5.9238	341.581

Therefore, the **3-period moving average** was selected as the most appropriate forecasting method for use as an input to capacity planning.

This result suggests that demand is **relatively stable with random variation**, rather than following a strong trend. Such behavior is consistent with **horizontal demand patterns**, as described in Chapter 9.

Forecast Output for Capacity Planning

Using the selected forecasting approach, the average forecasted demand is approximately **350–355 TON per month**. This forecast value is used as the **input demand (D)** in subsequent capacity planning calculations, including utilization analysis and capacity requirement estimation.

Role of Forecasting in Capacity Planning

Accurate demand forecasting enables the facility to:

- Estimate required processing hours
- Evaluate capacity utilization
- Identify potential capacity gaps
- Plan overtime or resource adjustments proactively

As emphasized in the course material, capacity planning without reliable demand forecasts would result in either **underutilized resources** or **excessive capacity strain**.

How the Facility Forecasts Demand and Evaluation of Forecasting Effectiveness

Demand Forecasting in the Selected Facility Evaluation of Forecasting Effectiveness

In the selected facility, demand forecasting is performed on a **monthly basis** and is primarily based on historical sales data. Forecasts are developed separately for retail products sold under the Nabil brand and for restaurant customers, which follow a make-to-order strategy.

Seasonal adjustments are applied to account for predictable demand increases during specific periods, such as Ramadan. Forecast results are used as direct inputs for production planning, material requirements planning (MRP), and capacity allocation.

The forecasting process is supported by the Odoo ERP system, which collects sales data, tracks historical demand, and enables planners to generate forecasts that are shared across production, procurement, and inventory management functions.

Evaluation of Forecasting Effectiveness

The current demand forecasting approach is effective for high-volume and standardized retail products, where historical demand patterns are relatively stable and predictable. Monthly forecasting allows the facility to plan production and procurement efficiently while maintaining acceptable service levels.

Additionally, the high product mix increases forecasting complexity and reduces overall forecast accuracy. While Odoo provides strong data integration and visibility, forecasting performance could be improved by using rolling forecasts, more frequent updates during peak seasons, and closer collaboration with key restaurant customers.

Capacity planning

What is Capacity Planning?

Capacity planning: Manufacturing capacity planning is a method makers use for calculating how much they can realistically produce on their production lines to keep up with forecasted demand. Capacity planning has to consider factors such as:

- Available resources
- Lead times

- Production processes

There capacity plan can be set to achieve short or long-term goals. Regardless of the period, In the selected facility, these factors are particularly important due to high product mix and long lead times.

the main objective of manufacturing capacity planning is to increase your profits and minimize costs.

Select the Capacity Measure

After forecasting demand, the next step in capacity planning is to determine **how capacity should be measured**. Selecting an appropriate capacity measure is essential because it defines the basis on which capacity availability, utilization, and requirements are evaluated.

Output Measures of Capacity

Output measures express capacity in terms of **finished units produced per unit of time**, such as units per day or kilograms per shiftoutput measures are best utilized when a firm produces a **small number of standardized products** with similar processing requirements.

In such environments, output measures provide a simple and intuitive way to evaluate capacity. However, when product variety is high and processing times differ significantly across products, output-based measures can become misleading and fail to reflect the true capacity of the system.

Input Measures of Capacity

Input measures define capacity in terms of **resources consumed**, such as:

- Machine hours
- Labor hours
- Work-center availability

The course material emphasizes that input measures are generally more suitable for **low-volume or flexible processes** where product mix and processing requirements vary. In these cases, capacity is constrained not by the number of finished units produced, but by the availability of time and resources.

Application to the Selected Facility

In the selected facility, the production line operates with a **high product mix**, producing more than thirty different products with varying weights, specifications, and processing times. In addition, production is split between retail products (make-to-stock) and restaurant orders (make-to-order), further increasing variability.

Due to these characteristics, using output measures of capacity (such as units per day) would not accurately represent system capability. Different products consume different amounts of processing time at critical operations, particularly at constrained resources such as the freezer and microwave.

input measures of capacity were selected as the most appropriate approach. Capacity is measured using **machine hours and labor hours**, which provide a consistent and comparable basis for evaluating resource availability across different products.

Estimate Capacity Requirements

The Data

$$\text{Utilization} = \frac{\text{Average output rate}}{\text{Maximum capacity}} \times 100\%$$

$$\text{Utilization} = \frac{1193.5}{1404} = 85\%$$

The capacity utilization is calculated by dividing the average output rate by the maximum capacity. In this case, the utilization is 85%, meaning that the system operates close to its maximum capacity while still maintaining a small buffer for variability.

- **Capacity cushion = 100% − Average Utilization rate (%)**

$$\boldsymbol{Capacity\ cushion = \%100 - \%85 = \%15}$$

Item	**Value**
Average demand (D)	**353.9 tons / month**
Annual demand	**4,200 tons / year**
Batch size	**1,200 kg (1.2 tons)**
Processing time (p)	**60 min = 1 hour per batch**
Critical resource	**Freezer**
Shifts	**2 shifts / day**

Operating time	**22 hours / day**
Utilization	**85%**
Capacity cushion (C)	**15%**

Step 2: Convert Data to Capacity-Planning Format

Annual Demand (D)

$$\boldsymbol{D = 353.9 \times 12 = 4,246 \textbf{ tons/year}}$$

Convert Demand to Number of Batches

$$\text{Number of batches per year} = \frac{4{,}246}{1.2} = 3{,}539 \text{ batches/year}$$

Processing Time (p)

Freezing time per batch:

$$p = 60 \text{ minutes} = 1 \text{ hour per batch}$$

Available Operating Hours per Year (N)

$$\boldsymbol{N = 22 \textbf{ hours/day} \times 365 \textbf{ days/year}}$$
$$\boldsymbol{N = 8,030 \textbf{ hours/year}}$$

Capacity Cushion (C)

$$\textbf{Utilization} = \boldsymbol{85\% \Rightarrow C = 15\%}$$

Apply the Capacity Requirement Formula

$$\boldsymbol{M = \frac{D \times p}{N\,[1 - (C/100)]}}$$

Substitute values:

$$M = \frac{3,539 \times 1}{8,030 \times (1 - 0.15)}$$

$$M = \frac{3,500}{6,825.5}$$

$$M = 0.518$$

Interpret the Result

What does M = 0.51 mean?

- The freezer requires 51% of one full-capacity unit
- One freezer is sufficient to meet the forecasted demand
- The system still maintains:
  - 15% capacity cushion
  - 85% utilization, consistent with the utilization calculation

This confirms that:

- The freezer is highly utilized
- But not overloaded
- Capacity is tight but stable, and any future increase in demand would directly affect the freezer

Explanation

Capacity requirements were estimated using input-based capacity measures, in line with the principles of capacity planning for high product-mix and batch based production systems. The analysis focused on the freezer operation, which was identified as the system bottleneck based on the Value Stream Map and its significantly longer processing time compared to other operations. Since overall system throughput is governed by the bottleneck resource, capacity calculations were performed at this stage rather than across the entire set of processing steps.

The average forecasted demand for the selected product is 350 tons per month, corresponding to 4,200 tons per year. Given a batch size of 1.2 tons and a freezing time of 60 minutes per batch, the total annual processing time required at the freezer was calculated to be 3,500 hours This value represents the total number of hours the freezer must be in operation annually to meet forecasted demand.

The facility operates two shifts per day, providing 22 operating hours per day which results in 8,030 available operating hours per year. To account for variability in demand, potential disruptions, and operational uncertainty, a capacity cushion of 15% was applied.

After applying the capacity cushion, the effective available freezer capacity is sufficient to meet the required processing hours. Using the capacity requirement formula the analysis indicates that the system requires approximately 51% of one freezer unit's full capacity. This result confirms that one freezer is currently sufficient to support the forecasted demand while maintaining a reasonable safety margin.

From a capacity-planning perspective, this finding suggests that the system is well balanced but tightly constrained. While there is no immediate capacity shortfall the freezer remains a critical resource and any future increase in demand reduction in operating hours, or increase in batch processing time would directly impact system performance. Therefore, continuous monitoring of freezer utilization and demand trends is essential to ensure long term operational stability and to support informed decisions regarding future capacity expansion or process improvement initiatives

Planning Material Requirements (MRP)

What is MRP ?

“Materials requirements planning (MRP) is probably the most rapidly growing system area in operations today. Hundreds of companies have implemented or are in the process of implementing an MRP system. Many more companies are evaluating the applicability of MRP for their operations.”

“MRP literature gives many impressive testimonials as to the results of successful implementation. Unfortunately, case histories of unsuccessful implementation efforts are also common. The potential MRP implementor may well be more discouraged by difficulties than encouraged by benefits.”

“Even in successful attempts, the implementation of an MRP system is a major undertaking for any company. Cost, both in resources and in time, is high. Additionally, since an MRP system interfaces with most functional areas, some degree of turmoil within the company is probably unavoidable

“Any company attempting or considering implementation of an MRP system needs information on how to manage the implementation process and where major problems can be expected to occur

“Since a materials requirements planning system demands an extremely high standard of accuracy of input data, record accuracy has frequently been mentioned as a major problem in the implementation of the syste

“The remaining problems in implementation are ‘people problems’. Generally, the survey results support this view, although technical problems such as forecasting demand were reported as major problems by some companies

Dependent vs. Independent Demand

After define what is the Material Requirements Planning (MRP)we should divide demand to independent and dependent demand. This step very important for of the MRP system and determines which items are forecasted and which are calculated.

Identify the End Item

The first action in this step is to clearly define the end item of the MRP system.

the selected end item is: Chicken Burger (1 kg pack)

The Chicken Burger is the final product go to customers by retail channels and restaurant client Its demand come directly from the market and is influence by customer orders seasonality and sales forecasts so the Chicken Burger is classified as an independent demand item.

Define Independent Demand

Independent demand refers to demand that:

- Comes directly from external customers
- Is not derived from another product
- Must be estimated using forecasting methods

In the Nabil Factory

- Demand for Chicken Burger is driven by:
    - Retail supermarket sales
    - Restaurant order
- This demand was previously forecasted using quantitative forecasting methods as we do in the back step we find the forecast by moving average methods

The forecasted monthly demand of 353 tons for Chicken Burger is an as independent demand as the primary input to the Master Production Schedule (MPS).

All Components Required to Produce the End Item

Once the independent demand item is defined, the next step is to list all raw materials and components required to manufacture one unit of the end item. These components are identified through the Bill of Materials (BOM).

For our product the main components include:

- Chicken meat
- Batter and spices
- Coating (tempura)
- Cooking oil
- Packaging materials (plastic bags and cartons)

Define Dependent Demand

Dependent demand come from :

- directly from the production of the end item
- Depends on the production quantity in the MPS
- Does not require forecasting

In our project all raw materials and packaging items required for producing Chicken Burger are classified as dependent demand item Their required quantities are calculated based on:

- The forecasted production quantity of Chicken Burger
- The quantities specified in the BOM

MRP Explosion

dependent demand items are not forecasted individually. Forecasting raw materials separately would introduce unnecessary uncertainty and lead to excess inventory or shortages.

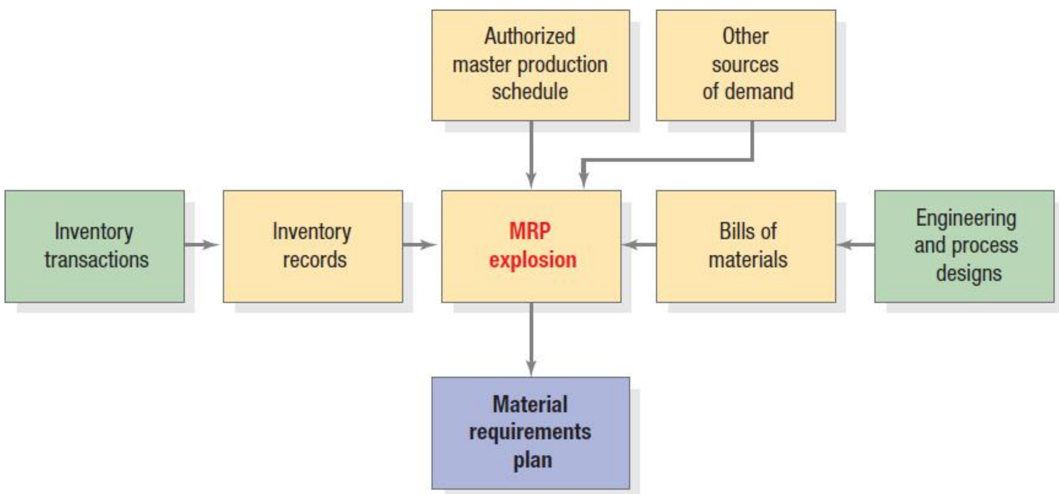


Figure 12 MRP Explosion Flow

MRP we use calculation process known as explosion, where:

- The independent demand (Chicken Burger) is expanded through the BOM
- Exact material requirements are generated for each component
- Material orders are timed according to lead times

In the Nabil Factory:

- Chicken Burger production volume are planned monthly based on forecasted demand.
- Raw material purchasing decisions are not based on forecasts but are calculated directly from production plans.
- Chicken meat, which has a long lead time, is planned carefully using safety stock (700 ton) and fixed order quantities to ensure supply continuity.

establishes the fundamental distinction between independent and dependent demand. In this project, the Chicken Burger is treated as the independent demand item, while all raw materials and packaging components represent dependent demand. This classification ensures that forecasting is applied only where appropriate and that material requirements are calculated accurately through the MRP system.

Bill Of Material -BOM-

In Material Requirements Planning (MRP), the **Bill of Materials (BOM)** defines the product structure by specifying all raw materials, components, and quantities required to produce one unit of the finished product. The BOM is a critical input to the MRP system because it enables the transformation of independent demand for the end item into dependent demand for raw materials.

In OUR project, the BOM was developed specifically for the Chicken Burger (1 kg pack) produced at Nabil Factory, ensuring full alignment between forecasting, capacity planning, and material planning.

End Item

- **Product name:** Chicken Burger
- **Unit of measure:** kg
- **Standard production unit:** 1 kg pack
- **Planning level:** Monthly

The Chicken Burger is the end item in the MRP system, and all materials listed in the BOM represent dependent demand.

Bill of Materials – Quantity per 1 Ton of Chicken Burger

Component	Quantity per 1 ton	Cost	Unit	Justification
Chicken meat	0.85	1325 JD	ton	Trimming losses, moisture loss, and yield reduction
Batter & spices	0.05	70JD	ton	Standard coating mix
Tempura / coating	0.06	40.55 JD	ton	Outer coating layer
Cooking oil	0.04	33 JD	L	Oil absorption during frying
Plastic Bags	1000	15 JD	كيس	
Carton Boxes	100	45 JD	كرتونه	

Table 1 BOM

Packaging Materials

- 1 kg Chicken Burger = 1 pack
- 1 ton = 1,000 packs
- Packaging assumptions:
    - 1 plastic bag per pack
    - 1 carton per 10 packs

MRP In Oddo

To support the Material Requirements Planning (MRP) process, the Bill of Materials (BOM) for the Chicken Burger product was implemented using the Odoo Manufacturing module. Odoo was selected as the ERP system because it integrates product data, inventory records, routing, and purchasing decisions within a single platform, enabling accurate material planning and execution.

Product and BOM Setup in Odoo

The finished product, Chicken Burger, was first defined in Odoo as a storable product with a unit of measure in kilograms. The standard production quantity was set to reflect the planning logic used in the MRP analysis, allowing material requirements to be calculated at the ton level. Once the product master data was created, a new Bill of Materials was linked to the Chicken Burger product, as illustrated in Figure .

The BOM specifies all raw materials and packaging components required to produce one ton of Chicken Burger, consistent with the assumptions defined in the analytical BOM. Each component was entered with its respective quantity and unit of measure, enabling Odoo to automatically explode material requirements based on the master production schedule.

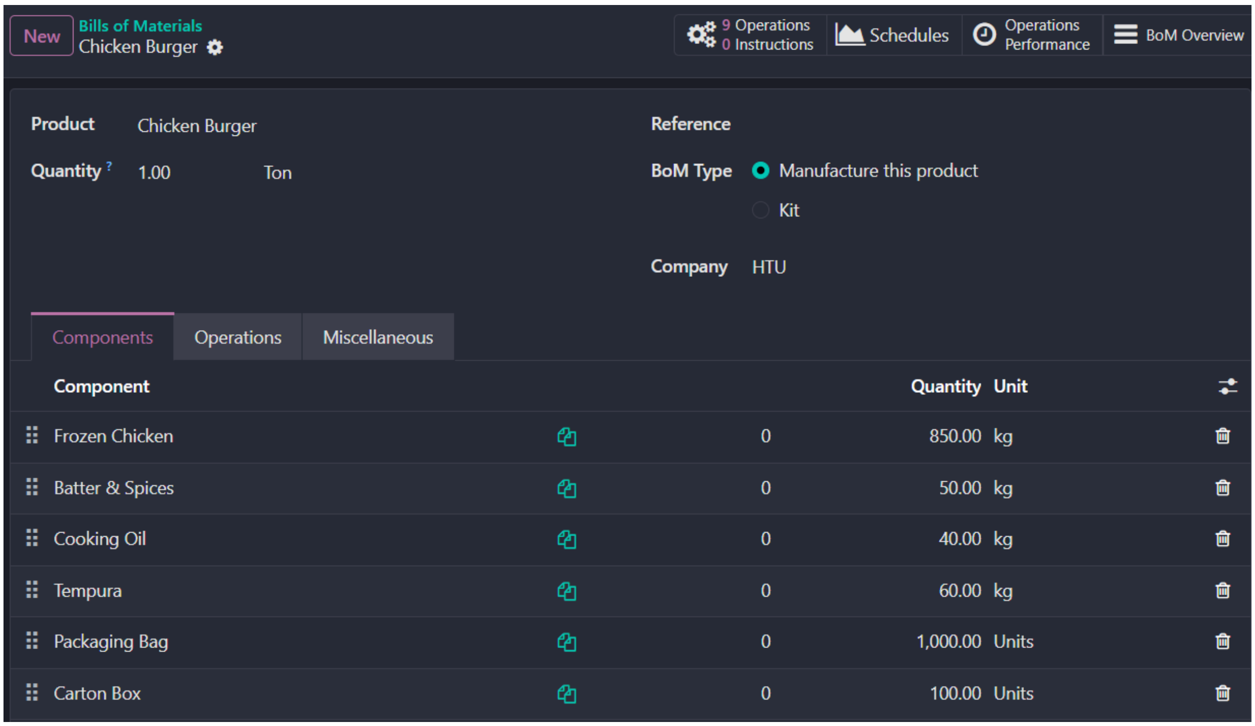


Figure 13.BOM IN ODDO

BOM Structure and Component Definition

Within Odoo, the BOM includes key raw materials such as chicken meat, batter and spices, tempura coating, and cooking oil, in addition to packaging materials including plastic bags and carton boxes. All components were defined as dependent-demand items, meaning their required quantities are calculated directly from the planned production quantity of the Chicken Burger rather than being forecasted individually

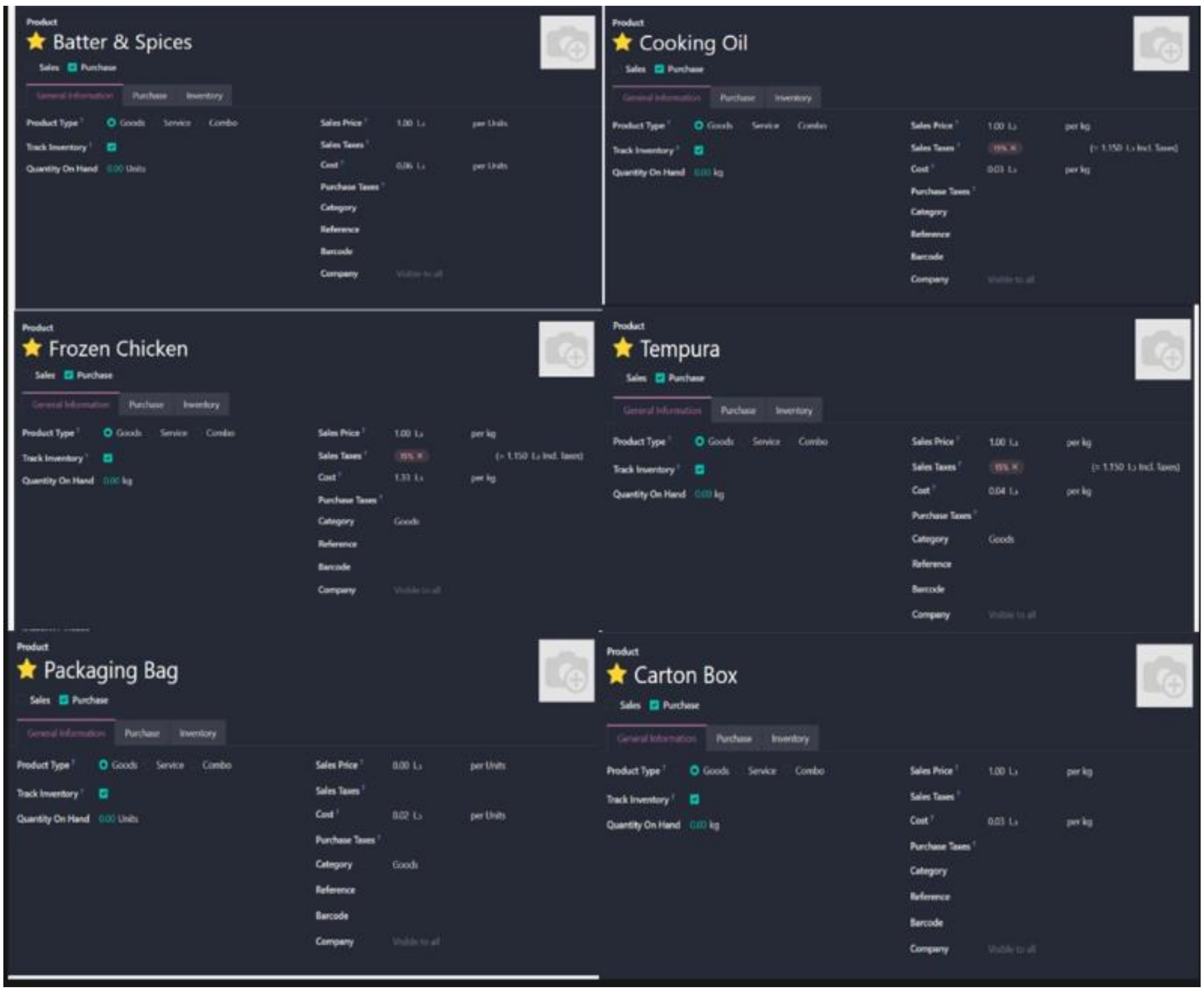


Figure 14 Component of BOM

This structure allows Odoo to translate independent demand for the finished product into precise, time-phased material requirements. The implemented BOM reflects realistic production yields and losses, ensuring that material planning aligns with actual manufacturing conditions at Nabil Factory.

Integration of Operations and Routing

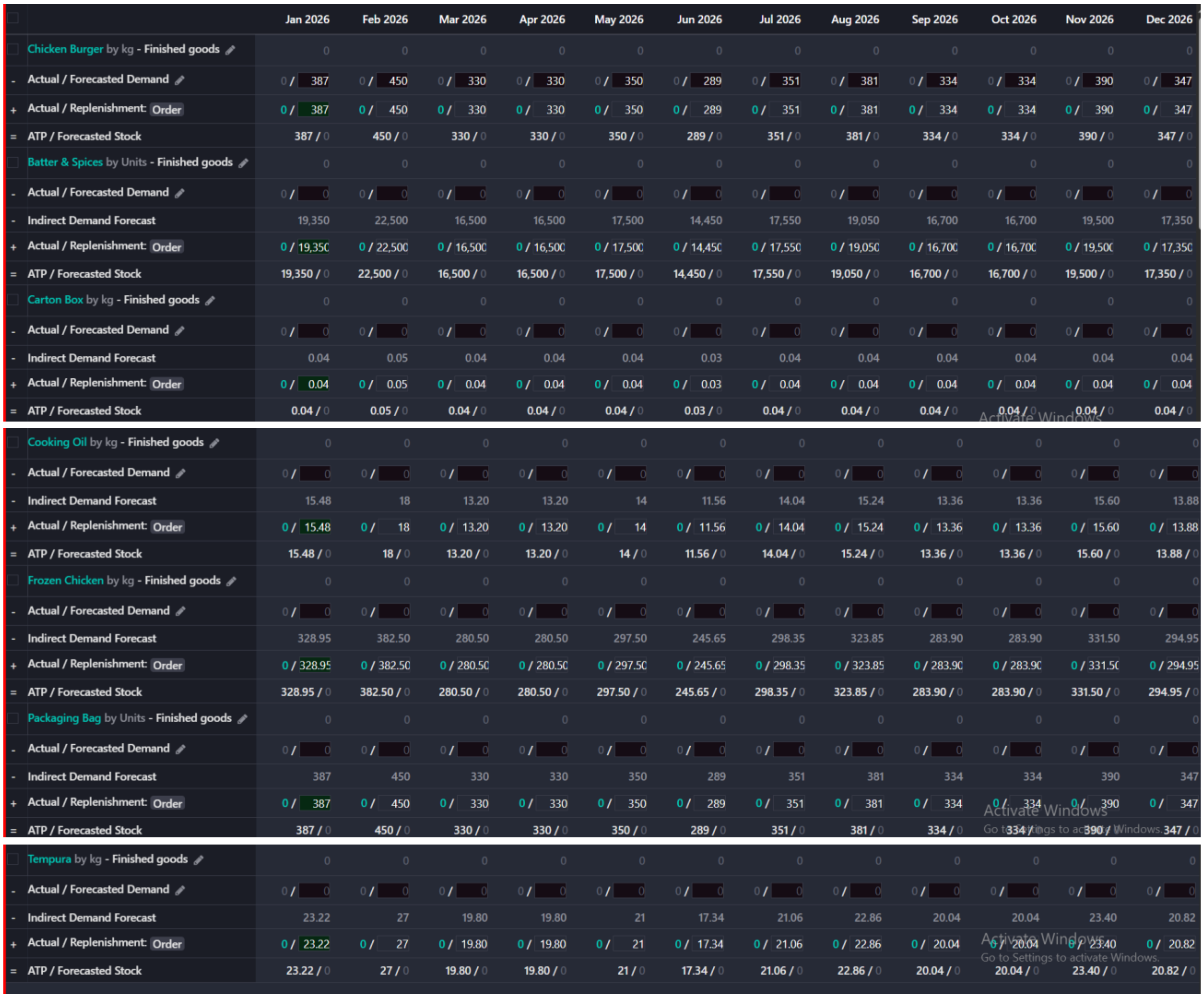
In addition to defining material components, production operations were configured within the BOM using Odoo's routing functionality. Each operation, such as mixing, forming, cooking, freezing, and packing, was assigned to a specific work center with a defined processing duration. As shown in *Figure* operation times were entered at the batch or ton level to maintain consistency with the MRP and capacity planning logic.

This integration ensures that Odoo considers both material availability and production capacity when generating manufacturing orders, supporting coordinated planning across the production line.

Operation	Work Center	Duration (minutes)
Microwave Heating	Microwave	05:00
Mixing	Mixing	15:00
Mechanical Forming	Forming	42:42
coating 1	Coating 1	48:00
coating 2	Coating 2	49:18
Frying	Frying	60:00
Cooking	Cooking	100:00
Freezing	Freezing	60:00
Packaging	Packaging	50:00

Role of the BOM in MRP Execution

Once the BOM is defined and validated, Odoo uses it as a core input for the MRP process. When a monthly production plan is released, the system automatically calculates required quantities of raw materials, checks inventory availability, and generates planned purchase orders based on lead times and lot-sizing rules. This functionality supports proactive procurement, reduces the risk of stockouts, and helps control inventory levels.



	Jan 2026	Feb 2026	Mar 2026	Apr 2026	May 2026	Jun 2026	Jul 2026	Aug 2026	Sep 2026	Oct 2026	Nov 2026	Dec 2026
Chicken Burger by kg - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 387	0 / 450	0 / 330	0 / 330	0 / 350	0 / 289	0 / 351	0 / 381	0 / 334	0 / 334	0 / 390	0 / 347
+ Actual / Replenishment: Order	0 / 387	0 / 450	0 / 330	0 / 330	0 / 350	0 / 289	0 / 351	0 / 381	0 / 334	0 / 334	0 / 390	0 / 347
= ATP / Forecasted Stock	387 / 0	450 / 0	330 / 0	330 / 0	350 / 0	289 / 0	351 / 0	381 / 0	334 / 0	334 / 0	390 / 0	347 / 0
Batter & Spices by Units - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
- Indirect Demand Forecast	19,350	22,500	16,500	16,500	17,500	14,450	17,550	19,050	16,700	16,700	19,500	17,350
+ Actual / Replenishment: Order	0 / 19,350	0 / 22,500	0 / 16,500	0 / 16,500	0 / 17,500	0 / 14,450	0 / 17,550	0 / 19,050	0 / 16,700	0 / 16,700	0 / 19,500	0 / 17,350
= ATP / Forecasted Stock	19,350 / 0	22,500 / 0	16,500 / 0	16,500 / 0	17,500 / 0	14,450 / 0	17,550 / 0	19,050 / 0	16,700 / 0	16,700 / 0	19,500 / 0	17,350 / 0
Carton Box by kg - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
- Indirect Demand Forecast	0.04	0.05	0.04	0.04	0.04	0.03	0.04	0.04	0.04	0.04	0.04	0.04
+ Actual / Replenishment: Order	0 / 0.04	0 / 0.05	0 / 0.04	0 / 0.04	0 / 0.04	0 / 0.03	0 / 0.04	0 / 0.04	0 / 0.04	0 / 0.04	0 / 0.04	0 / 0.04
= ATP / Forecasted Stock	0.04 / 0	0.05 / 0	0.04 / 0	0.04 / 0	0.04 / 0	0.03 / 0	0.04 / 0	0.04 / 0	0.04 / 0	0.04 / 0	0.04 / 0	0.04 / 0
Cooking Oil by kg - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
- Indirect Demand Forecast	15.48	18	13.20	13.20	14	11.56	14.04	15.24	13.36	13.36	15.60	13.88
+ Actual / Replenishment: Order	0 / 15.48	0 / 18	0 / 13.20	0 / 13.20	0 / 14	0 / 11.56	0 / 14.04	0 / 15.24	0 / 13.36	0 / 13.36	0 / 15.60	0 / 13.88
= ATP / Forecasted Stock	15.48 / 0	18 / 0	13.20 / 0	13.20 / 0	14 / 0	11.56 / 0	14.04 / 0	15.24 / 0	13.36 / 0	13.36 / 0	15.60 / 0	13.88 / 0
Frozen Chicken by kg - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
- Indirect Demand Forecast	328.95	382.50	280.50	280.50	297.50	245.65	298.35	323.85	283.90	283.90	331.50	294.95
+ Actual / Replenishment: Order	0 / 328.95	0 / 382.50	0 / 280.50	0 / 280.50	0 / 297.50	0 / 245.65	0 / 298.35	0 / 323.85	0 / 283.90	0 / 283.90	0 / 331.50	0 / 294.95
= ATP / Forecasted Stock	328.95 / 0	382.50 / 0	280.50 / 0	280.50 / 0	297.50 / 0	245.65 / 0	298.35 / 0	323.85 / 0	283.90 / 0	283.90 / 0	331.50 / 0	294.95 / 0
Packaging Bag by Units - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
- Indirect Demand Forecast	387	450	330	330	350	289	351	381	334	334	390	347
+ Actual / Replenishment: Order	0 / 387	0 / 450	0 / 330	0 / 330	0 / 350	0 / 289	0 / 351	0 / 381	0 / 334	0 / 334	0 / 390	0 / 347
= ATP / Forecasted Stock	387 / 0	450 / 0	330 / 0	330 / 0	350 / 0	289 / 0	351 / 0	381 / 0	334 / 0	334 / 0	390 / 0	347 / 0
Tempura by kg - Finished goods	0	0	0	0	0	0	0	0	0	0	0	0
- Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
- Indirect Demand Forecast	23.22	27	19.80	19.80	21	17.34	21.06	22.86	20.04	20.04	23.40	20.82
+ Actual / Replenishment: Order	0 / 23.22	0 / 27	0 / 19.80	0 / 19.80	0 / 21	0 / 17.34	0 / 21.06	0 / 22.86	0 / 20.04	0 / 20.04	0 / 23.40	0 / 20.82
= ATP / Forecasted Stock	23.22 / 0	27 / 0	19.80 / 0	19.80 / 0	21 / 0	17.34 / 0	21.06 / 0	22.86 / 0	20.04 / 0	20.04 / 0	23.40 / 0	20.82 / 0

Figure 15 .MPS

System Benefits and Alignment with Planning Objectives

The implementation of the Chicken Burger Bill of Materials (BOM) in Odoo plays a central role in linking demand forecasting, capacity planning, and material requirements planning at Nabil Factory. Demand forecasts are generated automatically from historical sales data in Odoo, particularly for restaurant orders, which represent a significant portion of production. These forecasts are then used as the starting point for production planning without the need for manual estimation, reducing planning errors and uncertainty.

Once the forecasted production quantity is defined, Odoo uses the Chicken Burger BOM to automatically calculate the exact quantities of raw materials and packaging components required for production. This includes chicken meat, batter and spices, oil, and packaging materials, all of which are planned based on the real production logic of one ton per manufacturing order.

In parallel, production capacity is considered through the definition of work centers and operation times In the implemented system the freezer is defined as a critical work center with a fixed processing time of 60 minutes per ton When manufacturing orders are generated Odoo clearly shows the workload on the freezer, making capacity constraints visible during planning This allows planners to recognize bottlenecks early and adjust production schedules accordingly, rather than reacting after delays occur

By combining accurate BOM data, real sales history, and defined work center capacities, Odoo enables effective coordination between production and purchasing activities. Purchasing decisions are no longer based on estimates but on system-generated requirements derived from confirmed manufacturing orders. This improves material availability, reduces the risk of shortages, and avoids unnecessary overstocking.

Overall, this integrated planning approach improves material visibility, supports better utilization of constrained resources such as the freezer, and creates a more stable and cost-effective production planning process for the Chicken Burger line at Nabil Factory

Inventory

Inventory Type

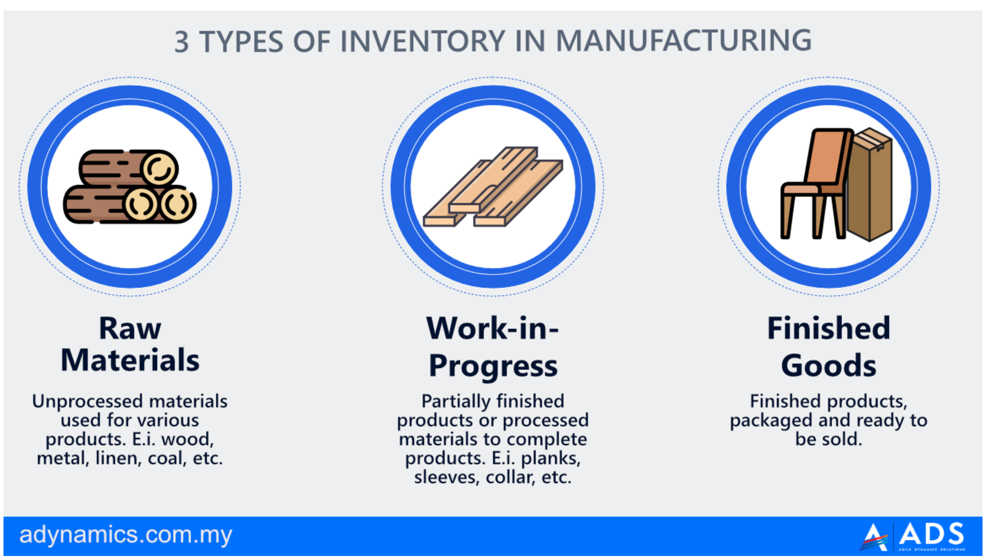


Figure 16 Inventory Type

Raw Material Inventory (Storage)

Raw material inventory, particularly frozen chicken, exists primarily to compensate for long and uncertain supplier lead times. As indicated in the Value Stream Map, a substantial portion of total lead time is attributable to upstream raw material inventory. From an operations planning perspective, this inventory is a deliberate decision made at the S&OP level to protect production capacity from supply risk.

Work-in-Process (WIP) Inventory

Work in process inventory accumulates between production stages as a result of uneven processing times and batch based scheduling. The freezer, which requires a fixed processing time of 60 minutes per ton as the system bottleneck., bottlenecks naturally create upstream WIP , WIP inventory is not a result of poor forecasting, but rather a structural outcome of capacity constraints and batch operations.

Finished Goods Inventory (Wherehouse)

Finished goods inventory supports the make-to-stock strategy used for **retail customers**. By holding packaged and frozen Chicken Burger products, Nabil Factory can respond quickly to stable demand patterns and seasonal peaks such as Ramadan. This inventory allows the firm to decouple production scheduling from immediate customer demand, which is a key concept emphasized in aggregate planning and scheduling

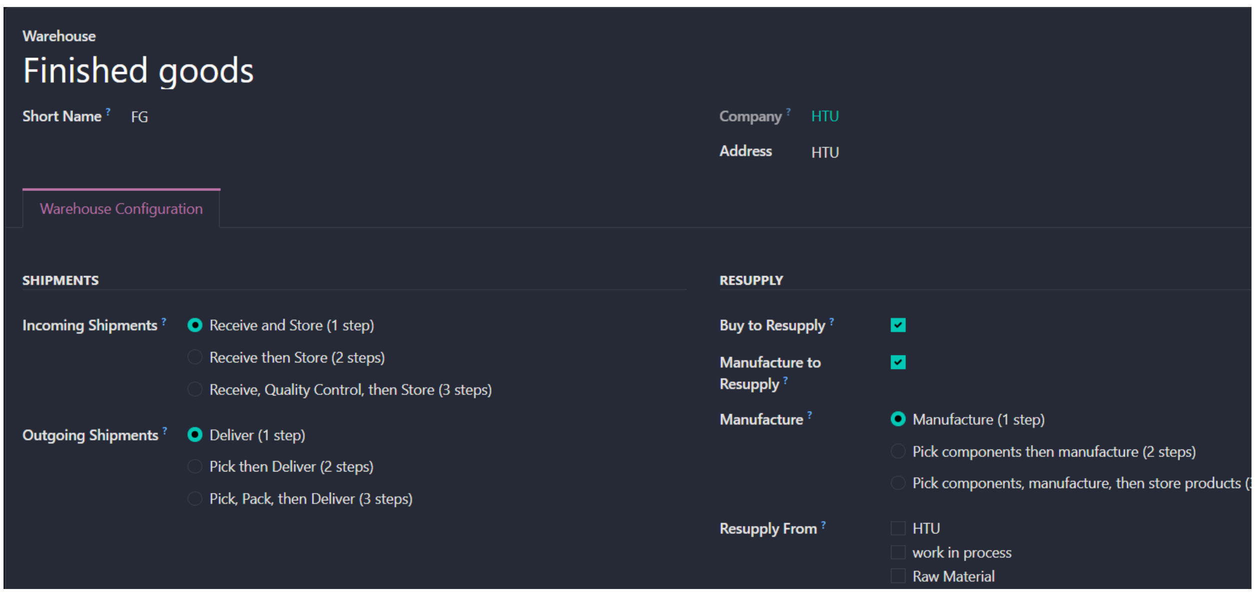


Figure 17 FG Wharehouse

How Inventory effect on Planning Implications

As highlighted in the lectures, inventory decisions involve trade-offs between service level and cost. Higher inventory levels improve customer responsiveness and reduce the risk of stockouts, but they also increase holding costs, storage requirements, and capital tied up in inventory. Lower inventory levels reduce costs but increase exposure to variability and operational risk.

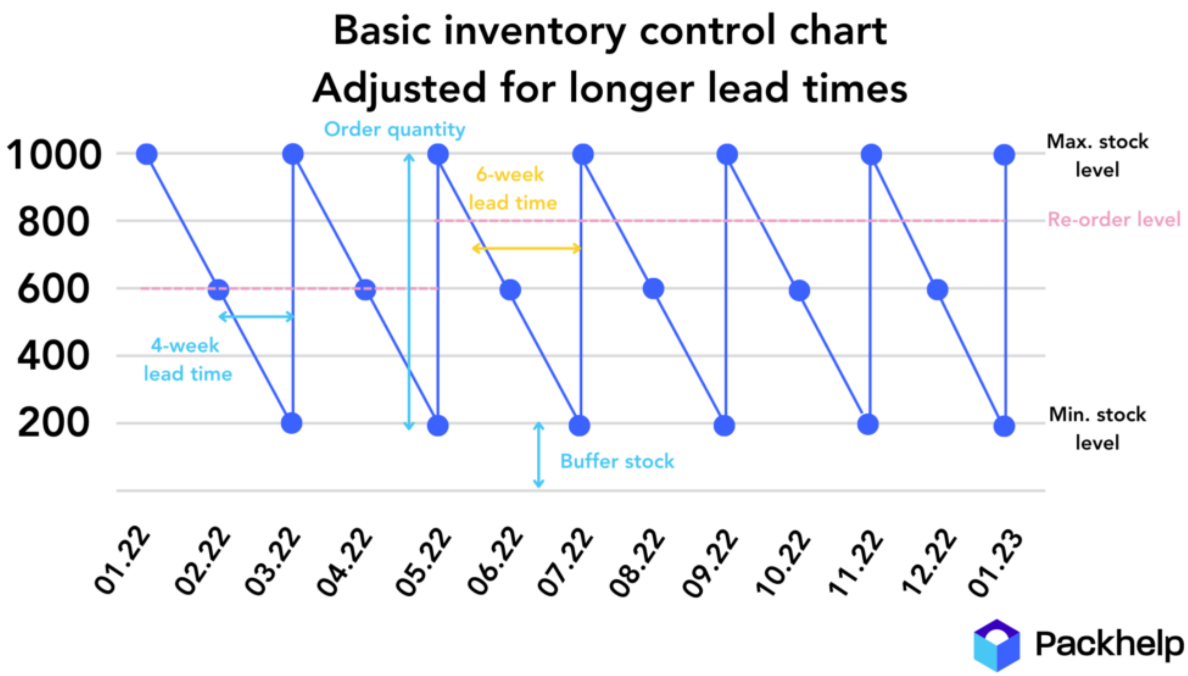


Figure 18 Inventory Buffer

At Nabil Factory, these trade-offs are clearly visible. While high raw material and WIP inventory ensure production stability, they also contribute to long lead times and low flow efficiency. From a planning perspective, improving supplier lead times and increasing bottleneck utilization would be more effective in reducing inventory than simply adjusting inventory targets.

System Support and Scheduling in Odoo

Inventory planning and scheduling at Nabil Factory are supported by the Odoo ERP system, which integrates inventory records, bills of materials, and manufacturing orderseffective scheduling requires accurate and timely information. Odoo provides real-time visibility of inventory levels and automatically checks material availability when manufacturing orders are released.

This integration ensures that scheduling decisions are consistent with inventory constraints and production plans. By linking inventory directly to MRP and scheduling, Odoo supports coordinated decision-making across planning levels and improves the overall effectiveness of production planning

Evaluation from an Operations Planning Perspective

Overall, inventory management at Nabil Factory is well aligned with the principles of operations planning and scheduling presented in the course. Inventory is not managed in isolation, but rather as an integral planning variable used to balance forecasted demand, available production capacity, and inherent system uncertainty. Through the integration of forecasting, master production scheduling, and material requirements planning, inventory decisions support stable production execution and reliable customer service, particularly for the Chicken Burger product.

The analysis shows that inventory levels at Nabil Factory are strongly influenced by structural characteristics of the production system In particular capacity bottleneck most notably the freezer operation and batch oriented scheduling decisions play a dominant role in determining both raw material and work in process inventory levels. These bottlenecks create natural points of accumulation upstream, leading to higher WIP inventory regardless of forecasting accuracy or inventory control policies. This outcome is consistent with the scheduling concepts discussed in the course, which emphasize that bottleneck resources dictate system flow and inventory behavior.

Furthermore, batch processing amplifies inventory accumulation by decoupling production stages and increasing waiting times between operations. While batch scheduling improves operational efficiency and food safety compliance, it also increases lead time and reduces flow efficiency. As a result, inventory at Nabil Factory serves not only as a buffer against demand variability but also as a compensating mechanism for capacity imbalances within the production line.

From an operations planning perspective, this analysis reinforces the importance of integrated planning across all levels. Efforts to reduce inventory solely through tighter inventory policies would have limited impact unless accompanied by improvements in bottleneck utilization, batch size optimization, or supplier lead-time reduction.

In conclusion, inventory performance at Nabil Factory reflects a rational and structured response to production constraints and market requirements. While inventory levels are higher than those observed in lean, continuous-flow systems, they are consistent with the operational realities of a batch-based food manufacturing environment. Continued improvements in integrated planning and bottleneck management would enable further reductions in inventory while maintaining system stability and service performance.

Scheduling

Definition of Scheduling

Scheduling refers to the detailed, short-term planning activity that allocates specific jobs to resources over time in order to execute the production planscheduling translates higher-level plans such as the Sales and Operations Plan and the production plan into actionable shop-floor decisions, including task sequencing, timing, and resource assignment.

At Nabil Factory scheduling represents the final planning level in the operations planning hierarchy and focuses on ensuring timely production of the Chicken Burger product while respecting capacity constraints and operational dependencies.

Scheduling Horizon and Planning Context

Scheduling at Nabil Factory is performed on a **weekly basis** which aligns with the planning framework where schedules operate over short time horizons such as days or weeks. Weekly scheduling provides a balance between operational stability and responsiveness, allowing the factory to meet delivery commitments while minimizing frequent rescheduling.

The Chicken Burger production line follows a **batch-based flow process**, meaning that products move through a fixed sequence of operations, but are processed in batches rather than continuously. This combination of batch and flow characteristics directly influences scheduling decisions, particularly with respect to work-in-process inventory and bottleneck utilization.

Process Flow and Routing Stability

In most cases, the Chicken Burger follows a standardized routing consisting of microwave, mixing, forming, battering, coating, frying, freezing, and packing. However, in some situations the cooking stage may be bypassed depending on product specifications. From a scheduling perspective, this introduces limited routing flexibility but does not fundamentally change the overall sequencing logic.

The routing structure is therefore considered **mostly stable**, which simplifies scheduling by allowing planners to focus on capacity constraints rather than frequent changes in process order.

Scheduling Rules and Job Sequencing

Given the high-volume and food-processing nature of the operation, scheduling at Nabil Factory is primarily governed by **First-In-First-Out (FIFO)** principles. FIFO sequencing supports fairness, traceability, and food safety requirements while reducing the risk of excessive waiting times for earlier batches.

All Chicken Burger orders—whether destined for retail or restaurant customers—are scheduled on the same production line using a common sequencing logic. Priority is not assigned at the individual order level; instead, delivery reliability is achieved through stable weekly schedules and the use of finished goods inventory for retail demand.

This approach reflects the scheduling concepts, where simple and robust dispatching rules are often preferred in batch and flow environments.

Bottleneck-Oriented Scheduling

The freezer operation, which requires **60 minutes per ton**, represents the primary bottleneck in the Chicken Burger production line. According to scheduling theory presented in the lectures, bottlenecks dictate system throughput and should therefore serve as the focal point of scheduling decisions.

At Nabil Factory, production is effectively paced by freezer availability. Upstream operations release batches based on freezer capacity to avoid excessive congestion, while downstream packing activities are scheduled to align with freezer output. This bottleneck-oriented scheduling approach helps stabilize flow and supports the goal of reducing overall lead time.

Release of Production and Use of Overtime

Production orders are released according to a fixed weekly plan, supported by the availability of raw materials and freezer capacity. Once the schedule is released, it remains largely fixed during execution, reflecting the high level of automation and process reliability in the facility.

To ensure adherence to delivery commitmentsNabil Factory employs **mandatory daily overtime** as a capacity adjustment mechanism overtime is a common short-term scheduling lever used to absorb variability without changing the underlying production plan. In this case, overtime supports delivery reliability while avoiding frequent rescheduling.

Scheduling Objectives and Performance Focus

The primary objectives of scheduling at Nabil Factory a

- **Meeting delivery deadlines**
- **Reducing overall production lead time**

These objectives are consistent with the scheduling performance measures discussed in the lectures, where due-date performance and flow time are critical indicators of scheduling effectiveness. By maintaining stable schedules and focusing on bottleneck utilization, the factory achieves reliable execution despite batch processing and long freezing times.

Scheduling Support through Odoo

Scheduling decisions are supported by the Odoo Manufacturing module, which translates manufacturing orders into planned operations with defined start and end times. While detailed real-time rescheduling is limited, Odoo provides visibility into operation sequences, work center assignments, and planned production timelines, enabling planners to monitor execution against the weekly schedule.

This system support ensures alignment between scheduling decisions and upstream planning activities such as MPS and MRP.

Evaluation of Scheduling Effectiveness

From an operations planning perspective, scheduling at Nabil Factory is effective in achieving delivery reliability and maintaining production stability. The use of weekly scheduling, FIFO sequencing, bottleneck-focused planning, and overtime as a capacity buffer aligns well with the principles However, batch processing and freezer constraints continue to drive work-in-process inventory and limit flow efficiency, indicating opportunities for future improvement through bottleneck enhancement rather than increased scheduling complexity.

Element	***Description***	
Scheduling horizon	Weekly	Short-term scheduling
Process type	Batch + Flow	Flow shop scheduling
Routing stability	Mostly fixed	Standardized routing
Dispatching rule	FIFO	Simple sequencing rules
Bottleneck	Freezer	Bottleneck scheduling
Capacity adjustment	Mandatory overtime	Short-term capacity control
Primary objective	On-time delivery & lead time reduction	Scheduling performance

Table 2Scheduling Characteristics of the Chicken Burger Production Line

Resource Allocation

Definition and Role in Production Planning

Resource allocation is the process of assigning available resources—such as labor, machines, and work centers—to production activities over time in a way that ensures demand is met while respecting capacity constraints and minimizing cost. Within production planning, resource allocation acts as the **execution bridge** between aggregate plans (forecasting and capacity planning) and detailed shop-floor scheduling.

At **Nabil Factory's Chicken Burger production line**, resource allocation is especially critical because multiple restaurant orders are processed simultaneously through **shared equipment**, while key operations—most notably freezing—operate under strict capacity and time constraints.

Workforce Allocation at the Aggregate Level

At the aggregate planning level, labor is the primary resource to be allocated. To evaluate alternative workforce allocation approaches, three strategies were analyzed: **Chase**, **Level**, and **Mixed** strategies.

The analysis showed that:

- The **Chase strategy** closely matches workforce size to demand but requires frequent hiring and layoffs, which increases operational instability and risk.
- The **Mixed strategy** combines hiring, layoffs, overtime, and undertime but resulted in the highest total cost.
- The **Level strategy** maintains a constant workforce and absorbs demand variability through overtime and undertime.

Based on total cost comparison and operational feasibility, the **Level strategy** was selected as the most appropriate workforce allocation approach:

- It achieved the **lowest total cost**
- It ensured workforce stability
- It aligned with food manufacturing requirements, where skill retention and process consistency are critical

This decision reflects a core production planning principle: **resource stability is often preferable to short-term efficiency when quality, safety, and reliability are priorities**.

How we deal with Operational Level

While the Level strategy defines *how many workers* are available, detailed resource allocation determines *how machines and operators are used* on the production line.

At Nabil Factory, all restaurant orders (A, B, and C) flow through the same sequence of operations:
Microwave → Mixing → Forming → Coating → Frying → Cooking → Freezing → Packing

These operations use **shared resources**, meaning that:

- A machine or operator can only process one job at a time
- Orders must wait when a resource is occupied
- Task start times are constrained by both technological precedence and resource availability

This allocation logic was implemented using finite-capacity scheduling, ensuring that resource conflicts are resolved realistically rather than assuming unlimited capacity.

Bottleneck-Oriented Resource Allocation

A key principle guiding resource allocation in this system is **bottleneck awareness**. The freezer was identified as the system's capacity-constrained resource due to:

- Long processing time per batch
- Batch-based operation
- Direct impact on total system throughput

As a result, resource allocation decisions were made to:

- Keep the freezer continuously utilized
- Allow upstream work-in-process buffers to form
- Avoid starving the bottleneck or overloading downstream operations

This approach prioritizes **system throughput over local efficiency**, which is consistent with modern production planning and constraint-based thinking.

Coordination Between Planning Level

Resource allocation at Nabil Factory is effective because it is **vertically aligned across planning levels**:

- **Forecasting** defines expected demand from restaurant contracts

- **Capacity planning** confirms feasibility and identifies constraints
- **Aggregate planning (Level strategy)** stabilizes labor resources
- **Scheduling** allocates shared machines and operators
- **Execution** uses overtime to absorb short-term variability

This coordination ensures that production plans are not only theoretically feasible but also practically executable on the shop floor.

Strengths

- Stable workforce with predictable skill availability
- Realistic allocation of shared machines
- Bottleneck-focused utilization of critical resources
- Strong delivery reliability for restaurant customers

Limitations

- Dependence on overtime to handle demand peaks
- Accumulation of work-in-process before freezing
- Limited flexibility to respond to sudden demand shocks

Overall, resource allocation at Nabil Factory is **well controlled and aligned with production planning principles**, but it relies on buffers and overtime to compensate for structural constraints rather than eliminating them.

- A **Level workforce strategy** at the aggregate planning level
- **Finite-capacity scheduling** at the operational level
- **Bottleneck-oriented execution** at the system level

Cost Optimization

Why we do Cost Optimization

Cost optimization at Nabil Factory does not aim to reduce costs by cutting quality, safety, or delivery performance.
Instead, it focuses on reducing avoidable operating costs that arise from system inefficiencies, particularly those related to:

- Excess inventory

- Overtime dependency
- Quality losses
- Long lead times
- Batch-driven flow inefficiencies

Given the strict delivery requirements of restaurant customers and food safety constraints, cost optimization must be achieved **through system-level improvements**, not local cost cutting.

Identification of Major Cost Drivers

Based on the Value Stream Map, Capacity Planning, OEE, Inventory, and Scheduling analyses, the **main cost drivers** in the Chicken Burger system are:

1. Inventory holding cost
2. Overtime labor cost
3. Quality loss (scrap and waste)
4. Indirect flow-related costs (WIP, waiting, energy)

3. Inventory Holding Cost Analysis

1-- Current Inventory Situation

From Nabil data :

- Average monthly demand = **350 tons**
- Raw material safety stock = **700 tons**
- Supplier lead time ≈ **2 months**
- Raw material = frozen chicken (high storage cost)

This means:

Raw material inventory alone covers **≈2 months of demand**

2-- Estimating Inventory Holding Cost

we use **conservative industry assumptions**

- Average cost of frozen chicken ≈ **1,325 JD / ton** --from BOM)
- Annual holding cost rate (cold storage + capital + energy) ≈ **15%**

Inventory Value

$$700 \text{ tons} \times 1{,}325 \text{ JD/ton} = 927{,}500 \text{ JD}$$

Annual Holding Cost

$$927{,}500 \times 0.15 = \mathbf{139{,}125} \text{ JD/year}$$

3-- Optimization Opportunity (From Futur- VSM)

From **my analyze on Future - VSM**:

- Supplier lead time reduced from **2 months → 2 weeks**
- This allows raw material inventory to drop safely by **≈50%**

New Inventory Level

$$700 \times 0.5 = 350 \text{ tons}$$

New Holding Cost

$$463{,}750 \times 0.15 = 69{,}563 \text{ JD/year}$$

Annual Saving from Inventory Optimization

$$139{,}125 - 69{,}563 = \mathbf{69{,}562} \text{ JD/year}$$

This saving is achieved **without increasing risk**, because it is enabled by lead-time reduction, not aggressive inventory cuts.

Overtime Cost Analysis

1 Current Overtime Practice

- Mandatory overtime used **daily**
- Overtime is the primary buffer for:
    - Demand variability
    - Scheduling slippage
    - Batch congestion

This indicates **structural reliance on overtime**, not occasional use.

2 Estimating Overtime Cost

Conservative assumptions

- Average labor cost ≈ **5 JD/hour**
- Overtime premium = **1.5×**
- Extra overtime ≈ **2 hours/day**
- Estimated affected workforce ≈ **20 operators**

Daily Overtime Cost

$$20 \times 2 \times (5 \times 1.5) = 300 \text{ JD/day}$$

Annual Overtime Cost

$$300 \times 300 \text{ working days} = \mathbf{90,000} \text{ JD/year}$$

Optimization Opportunity

Controlled capacity cushion (15%)

- Bottleneck-paced scheduling
- Reduced unplanned congestion

Conservative expected reduction:

20–25% reduction in overtime

Annual Saving

$$90{,}000 \times 0.25 = \mathbf{22,500} \text{ JD/year}$$

The system does **not lack capacity**; it lacks **flow coordination**.
Cost reduction comes from planning discipline, not adding machines.

Quality Cost

1 --- Current Quality Loss

- Waste rate = **12%**
- Annual production ≈ **4,200 tons**

- Waste quantity:

$$4{,}200 \times 0.12 = 504 \text{ tons/year}$$

2 -- Cost of Waste

Using conservative **raw material cost only**:

$$504 \times 1{,}325 = \mathbf{667{,}800} \text{ JD/year}$$

3 Optimization Opportunity by Six Sigma – DMAIC

Waste Reduction through DMAIC Application

Waste reduction in the Chicken Burger production line at Nabil Factory was achieved through the structured application of the **DMAIC methodology**, which enabled a systematic reduction of defects and process variability. The analysis identified waste as a critical issue, with an observed rate of approximately **8%**, primarily occurring at the **oil frying and cooking stages**, where products cannot be reworked once defects occur.

During the **Define** phase, these stages were classified as critical-to-quality processes due to their direct impact on product conformity and cost. In the **Measure** phase, waste was quantified using production and quality data, showing annual waste of approximately **336 tons**. This converted waste from a qualitative problem into a measurable performance metric.

The **Analyze** phase revealed that waste was driven mainly by process variability rather than isolated operator errors. Key root causes included inconsistent raw chicken quality, temperature fluctuations during frying, and production pressure caused by batch processing and tight schedules.

In the **Improve** phase, targeted actions were implemented, including tighter temperature control, standardized cooking parameters, improved supplier selection, and better alignment of production flow with the system bottleneck. These measures reduced variability at the most sensitive stages and lowered the waste rate from **12% to approximately 5%**.

Finally, the **Control** phase ensured sustainability through continuous monitoring of defect rates and integration of quality indicators into daily production reviews. Overall, DMAIC enabled waste reduction by addressing root causes systematically, leading to lower material losses, improved process stability, and significant cost savings without compromising food safety or delivery reliability.

New Waste Quantity

$$4{,}200 \times 0.05 = 210 \text{ tons}$$

New Waste Cost

$$210 \times 1{,}325 = 278{,}250 \text{ JD}$$

Annual Saving from Quality Improvement

$$445{,}200 - 278{,}250 = \mathbf{166{,}950} \text{ JD/year}$$

This is the **largest cost lever** in the system.

Indirect Flow-Related Costs

These costs are harder to measure precisely, but their **existence is proven** by:

- Activity ratio = **0.12%**
- Extremely long lead time vs processing time
- Excess freezer waiting and WIP

From literature and food-industry benchmarks:

A **10–15% reduction in WIP** typically yields **5–8% reduction in indirect operating costs**

Conservative estimate:

- Indirect cost base ≈ **100,000 JD/year**
- Saving ≈ **7%**

Estimated Saving

$$\mathbf{7{,}000} \text{ JD/year}$$

Total Quantified Cost Optimization Impact

Cost Category	Annual Saving (JD)
Inventory holding	69,562
Overtime	22,500

Cost Category	Annual Saving (JD)
Quality waste	166,950
Indirect flow costs	7,000
Total Estimated Saving	**≈266,000 JD/year**

Critical Interpretation

From a systems engineering perspective, these savings are **not achieved by cost cutting**, but by:

- Reducing lead time (inventory)
- Reducing variability (quality)
- Improving flow synchronization (overtime & WIP)
- Managing constraints instead of compensating for them

Importantly:

- Delivery reliability is **maintained**
- Food safety is **not compromised**
- No capital investment is required

Cost optimization at Nabil Factory is driven primarily by system inefficiencies rather than insufficient capacity or poor execution. Quantitative analysis shows that excessive inventory, high overtime dependence, and quality losses represent the largest sources of avoidable cost. By reducing supplier lead time, improving bottleneck coordination, stabilizing process quality, and controlling overtime usage, the Chicken Burger production line can achieve annual cost savings of approximately 266,000 JD. These improvements do not rely on workforce reduction or capacity expansion, but instead reflect a mature application of manufacturing systems engineering principles focused on flow efficiency, variability reduction, and constraint management.

Manufacturing Approach

The Chicken Burger production at Nabil Factory follows a hybrid manufacturing approach, with a strong dominance of Make-to-Order (MTO) for restaurant customers, supported by limited Make-to-Stock (MTS) elements to ensure system stability.

Primary Manufacturing Approach: Make-to-Order (MTO)

For restaurant customers, the Chicken Burger is primarily produced under a Make-to-Order strategy. Although Nabil Factory operates under fixed monthly contracts with major restaurant chains, production is still triggered by confirmed customer orders rather than purely speculative forecasts. Each restaurant places a minimum committed order of approximately 300 tons per month, which creates a stable but order-driven demand structure.

This approach aligns with MTO principles as presented in the course material, where production begins after demand is known, and the product is tailored to customer requirements. In this case, each restaurant specifies unique product characteristics such as patty size, weight, thickness, and sometimes processing steps (for example, partial processing without final cooking). These customer-specific requirements make full standardization impossible and justify the use of an MTO approach rather than a pure make-to-stock system.

From an operations planning perspective MTO allows Nabil Factory to:

- Reduce the risk of producing non-conforming products
- Avoid excessive finished goods inventory for customized items
- Maintain strong alignment between customer contracts and production plans

However, the MTO approach also increases system sensitivity to scheduling accuracy and bottleneck utilization, especially at batch-based operations such as freezing.

Supporting Manufacturing Logic: Make-to-Stock (MTS) Elements

Despite the dominant MTO structure, the system also incorporates Make-to-Stock elements. Nabil Factory maintains a safety stock of approximately 700 tons of frozen Chicken Burger products. This inventory serves as a buffer against demand variability, production disruptions, and supplier lead-time uncertainty.

This hybrid structure reflects the reality of food manufacturing systems where pure MTO is often impractical due to food safety constraints, long processing times, and capacity limitations. The freezer bottleneck, identified in the Value Stream Map, makes it operationally risky to rely exclusively on immediate production. Therefore, safety stock supports service level reliability while production remains contract-driven.

Engineer-to-Order (ETO): Not Applicable

The Engineer-to-Order (ETO) approach is not suitable for the Chicken Burger production line. While minor adjustments such as changing forming molds are possible, the core product design, recipe, and processing technology remain standardized. No new product engineering or unique process design is developed for individual customers, which clearly excludes ETO as a viable manufacturing strategy.

This aligns with course concepts that define ETO as appropriate only when products require unique engineering design, which is not the case in high-volume food manufacturing.

MTO Vs MTS Vs ETO

Manufacturing Approach Evaluation

From a critical operations management perspective, the hybrid Make-to-Order (MTO)–Make-to-Stock (MTS) manufacturing approach adopted by Nabil Factory for the Chicken Burger product is structurally justified, yet it also exposes several inherent system limitations. The presence of fixed monthly contracts with restaurant customers significantly reduces demand uncertainty compared to open-market forecasting. However, this stability at the demand level does not fully translate into operational flexibility due to internal process constraints.

As identified through the Value Stream Mapping analysis, the production system is heavily influenced by batch processing and a dominant freezer bottleneck ans production is initiated based on confirmed orders (MTO logic)the batch-oriented nature of upstream and downstream processes forces the system to behave like a push system particularly before and at the freezing stage. This mismatch between order-driven demand and batch-driven execution leads to extended lead times and substantial work-in-process accumulation.

To maintain delivery reliability under these conditions, the system relies extensively on three compensating mechanisms. First, mandatory overtime is used as a short-term capacity adjustment tool to absorb fluctuations in demand and recover from schedule deviations. While overtime provides immediate flexibility, it also increases labor costs and may negatively impact long-term workforce sustainability. Second, inventory buffers, particularly in raw materials and finished goods, are maintained to decouple production from demand variability and supplier lead-time uncertainty also this supports service-level performance, it significantly increases holding costs and ties up working capital Third push based scheduling upstream of the freezer is applied to ensure continuous utilization of the bottleneck resource while this maximizes throughput at the freezer, it also results in upstream congestion and low flow efficiency, as reflected by the very low activity ratio observed in the current-state VSM.

From a flow perspective, this hybrid approach prioritizes delivery reliability over flow efficiency. The system is optimized to ensure that customer commitments are m even if this requires longer internal lead times, higher inventory levels, and increased operational costs. This trade-off is consistent with operations management theory, which recognizes that systems constrained by batch processes and bottlenecks often sacrifice flow efficiency to maintain service performance.

However, the analysis also reveals opportunities for improvement within the existing manufacturing approach. Introducing a more pull-oriented scheduling mechanism downstream of the forming process could reduce unnecessary WIP accumulation by better synchronizing production release with actual bottleneck capacity. Additionally, targeted improvements in freezer utilization, such as improved batch sequencing or thermal pre-conditioning upstream, could increase effective capacity without major capital investment. These measures would allow the system to retain the benefits of the MTO–MTS hybrid model while mitigating its most significant inefficiencies.

Inventory Control Logic: P-System versus Q-System

The inventory control policy at Nabil Factory closely resembles a P-system (periodic review system) rather than a Q-system (continuous review system). In a P-system, inventory levels and replenishment decisions are reviewed at fixed, predetermined intervals, whereas a Q-system triggers replenishment whenever inventory falls below a specified reorder point. At Nabil Factory, both production planning and procurement decisions are reviewed on a monthly basis, consistent with the contract cycles established with restaurant customers and the Sales and Operations Planning (S&OP).

This periodic review structure aligns naturally with the hybrid MTO–MTS manufacturing approach. Monthly contracts provide a stable planning signal, enabling the firm to synchronize inventory reviews, production planning, and capacity allocation within a single planning cyclethe   Q-system would require continuous monitoring and rapid replenishment, which would be incompatible with long supplier lead times, batch freezing requirements, and limited short-term capacity flexibility.

While the P-system supports planning stability and coordination, it also reduces responsiveness to short-term demand changes and increases reliance on safety stock. This reinforces the system’s dependence on inventory buffers and overtime as primary riskmitigation tools. Therefore, the choice of a P-system is consistent with the current manufacturing strategy, but it also amplifies the cost and lead time trad inherent in batch based bottleneck constrained production

Evaluation of the Effectiveness of Production Planning Methods at Nabil Factory

The effectiveness of production planning at Nabil Factory can be evaluated by analyzing how well forecasting, aggregate planning, capacity planning, material planning, and scheduling work together to achieve operational objective the planning system can be described as effective and reliable, particularly in terms of delivery performance and system stability, but it also shows clear signs of compensating for structural inefficiencies rather than eliminating them.

Planning Area	***Method Used***	***Effectiveness***	***Strengths***	***Limitations***
Demand Planning	Monthly contracts	High	Stable demand, predictable planning	Limited flexibility to shocks
S&OP	Monthly aggregate planning	Medium–High	Clear production targets	Relies on overtime
Capacity Planning	Installed capacity + overtime	Medium	Sufficient on average	Tight capacity, low resilience
Inventory Planning	Safety stock (P-system)	High	High availability	Long lead time, high holding cost
Scheduling	Weekly, batch-based	Medium	Stable execution	Long WIP, limited pull
System Integration	Odoo ERP	Medium–High	Good visibility	Manual decisions remain

Table 4 Evaluation of Production Planning Methods at Nabil Factory

Demand Forecasting and Planning Effectiveness

Sales and Operations Planning

Nabil Factory operates with a clear monthly production plan that covers the entire facility rather than individual production lines. This centralized planning approach supports coordination across departments and simplifies decision making this plan process is effective in aligning demand commitments with available resources at a high level.

However, when happen gap between planned and actual performance, the primary corrective action is the use of overtime rather than revising the production plan While this maintains schedule stability it also indicates limited adaptability within the planning system The S&OP process is therefore effective in maintaining control and predictability, but less effective in enabling proactive adjustments.

Capacity Planning Effectiveness

apacity planning at Nabil Factory can be described as **adequate but tight**, meaning that the factory is generally able to meet customer demand, but with little margin for error. The existing production capacity is sufficient under normal operating conditions, and key resources specially the freezerare designed to handle large batch sizes. This allows the factory to process high volumes of Chicken Burger in a single cycle, which supports stable production output.

So this large batch capacity the freezer is usually **highly utilized** but it does not always directly cause production delays the system can keep moving as long as production follows the planned schedule. However, the high level of utilization also means that there is limited spare capacity available to see unexpected disruptions such as demand fluctuations or minor process delays

To compensate for this lack of flexibility, Nabil Factory relies heavily on **overtime** as a short-term capacity adjustment tool. Overtime allows the factory to maintain delivery commitments when demand increases or when production falls behind schedule. While this approach is effective in the short term, it also increases labor costs and places additional pressure on the workforce.

From a production planning perspective, this indicates that capacity planning at Nabil Factory focuses more on **reacting to variability** Although the installed capacity is sufficient on average, the system depends on additional labor input to remain stable. A more detailed analysis of capacity utilization and workload distribution across production stages could help reduce the reliance on overtime. Improving load balancing and increasing flexibility at non bottleneck operations would make the system more resilient and improve overall planning performance.

Material Requirements Planning (MRP) and Inventory Planning

Material planning at Nabil Factory is largely manual, even though it is supported by Odoo. Raw materials—especially chicken meat—are always available because procurement is done in large quantities with long lead times of up to two months. The 700-ton safety stock is justified given the average monthly demand of 350 tons and long replenishment cycles.

This approach ensures high production continuity and eliminates the risk of material-related stoppages SO it also increases inventory holding costs and system lead time. Inventory planning is therefore effective in ensuring availability, but it is not optimized for flow efficiency or cost minimization.

The planning logic closely resembles a P system (periodic system), with monthly review cycles aligned with S&OP and contract structures. This reinforces planning stability but reduces responsiveness to short-term changes.

		Jan 2026	Feb 2026	Mar 2026	Apr 2026	May 2026	Jun 2026	Jul 2026	Aug 2026	Sep 2026
	Chicken Burger by kg - Finished goods	0	3	0	0	0	0	0	0	0
-	Actual / Forecasted Demand	0 / 387	0 / 450	0 / 330	0 / 330	0 / 350	0 / 289	0 / 351	0 / 381	0 / 334
+	Actual / Replenishment: Order	390 / 387	0 / 447	0 / 330	0 / 330	0 / 350	0 / 289	0 / 351	0 / 381	0 / 334
=	ATP / Forecasted Stock	390 / 3	450 / 0	330 / 0	330 / 0	350 / 0	289 / 0	351 / 0	381 / 0	334 / 0
	Batter & Spices by Units - Finished goods	0	0	0	0	0	0	0	0	0
-	Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
-	Indirect Demand Forecast	19,350	22,350	16,500	16,500	17,500	14,450	17,550	19,050	16,700
+	Actual / Replenishment: Order	0 / 19,350	0 / 22,350	0 / 16,500	0 / 16,500	0 / 17,500	0 / 14,450	0 / 17,550	0 / 19,050	0 / 16,700
=	ATP / Forecasted Stock	19,350 / 0	22,350 / 0	16,500 / 0	16,500 / 0	17,500 / 0	14,450 / 0	17,550 / 0	19,050 / 0	16,700 / 0
	Carton Box by kg - Finished goods	0	0	0	0	0	0	0	0	0
-	Actual / Forecasted Demand	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
-	Indirect Demand Forecast	0.04	0.05	0.04	0.04	0.04	0.03	0.04	0.04	0.04
+	Actual / Replenishment: Order	0 / 0.04	0 / 0.05	0 / 0.04	0 / 0.04	0 / 0.04	0 / 0.03	0 / 0.04	0 / 0.04	0 / 0.04
=	ATP / Forecasted Stock	0.04 / 0	0.05 / 0	0.04 / 0	0.04 / 0	0.04 / 0	0.03 / 0	0.04 / 0	0.04 / 0	0.04 / 0
	Cooking Oil by kg - Finished goods	0	0	0	0	0	0	0	0	0

Figure 19 Manufacturing Order and Scheduled Operations for Chicken Burger

Scheduling Effectiveness

Scheduling at Nabil Factory is largely stable and fixed, typically performed on a weekly basis. The production line operates continuously, with minimal visible downtime, indicating effective coordination between operations. Waiting and WIP are present but managed through batching and continuous line operation rather than frequent rescheduling.

When scheduling problems arise, they are addressed through a combination of overtime and inventory usage, rather than schedule redesign. This reinforces delivery reliability but masks underlying flow inefficiencies. From a scheduling theory perspective, the system prioritizes on-time delivery over lead-time reduction, which is consistent with batch-based manufacturing environments.

Planning Decision	Impact on Lead Time	Impact on Cost	Impact on Service Level
Batch processing	Increases	Neutral	Positive
Overtime usage	Neutral	Increases	Positive
Safety stock	Increases	Increases	Positive
Bottleneck focus	Neutral	Neutral	Positive
Push-based planning	Increases	Increases	Neutral

Table 5 Impact on Performance

OVERALL Evaluation

om a critical operations management standpoint, production planning at Nabil Factory can be considered **effective in achieving its short-term and operational objectives**, particularly delivery reliability, production continuity, and overall system stability. The planning system demonstrates strong internal coordination by integrating fixed customer contracts, inventory buffers, overtime, and batch-based production logic into a coherent planning framework. This integration ensures that customer commitments are consistently met, even under conditions of demand variability and long supplier lead times.

However, when evaluated beyond immediate execution performance, the effectiveness of the planning system reveals **significant structural limitations**. Rather than actively optimizing system flow, the current planning approach relies on **compensatory**

mechanisms to manage underlying inefficiencies. Long lead times persist not because they are unavoidable, but because batch processing and inflexible upstream supply structures are accepted as fixed constraints. Inventory is used extensively as a buffer to absorb uncertainty, which masks problems related to flow imbalance and capacity synchronization instead of resolving them.

The heavy dependence on **overtime** further highlights this reactive nature of planning. While overtime provides short-term flexibility and supports delivery reliability, it indicates that capacity planning lacks sufficient robustness to handle variability within normal operating conditions. From a critical perspective, overtime functions as a corrective tool rather than a planned capacity strategy, increasing operational costs and reducing long-term sustainability.

Moreover, the system's limited responsiveness to unexpected market changes such as sudden demand drops or external disruptions suggests that planning effectiveness is highly contingent on stable conditions. The reliance on periodic review cycles and push-based production upstream reduces the system's ability to adjust quickly without resorting to additional buffers or labor input. This indicates that the planning system prioritizes **control and predictability over adaptability and flow efficiency**.

In essence, production planning at Nabil Factory is **well-controlled but not fully optimized**. It succeeds by managing variability through inventory, overtime, and batching, rather than by redesigning processes to reduce variability at its source. While this approach is operationally pragmatic and common in batch-oriented food manufacturing, it limits the system's potential for lead-time reduction, cost efficiency, and lean performance. A more optimized planning system would shift from compensating for structural constraints toward systematically reducing their impact through improved flow synchronization, bottleneck-focused capacity enhancement, and more responsive planning mechanisms.

However, this effectiveness comes at the cost of:

- Long lead times
- High inventory levels
- Dependence on overtime
- Limited responsiveness to unexpected market changes

Strengths	Weaknesses
Stable contracts	Long lead time

High service level	High inventory
Strong execution	Overtime dependency
Integrated ERP	Limited pull

Table 6 Strengths and Weaknesses of Production Planning

In other words, the planning system is well-controlled but not fully optimized. It compensates for structural constraints—such as batch processing and long supplier lead times—rather than eliminating them.

Item	Planned	Actual	Observation
Monthly output	350 tons	350–370 tons	Overtime used
Working hours	22 hrs/day	24+ hrs/day	Mandatory OT
Inventory level	700 tons	750 tons	Buffer increase

Table 7 Planned vs Actual Production

How Planning Techniques Help Identify the Appropriate Manufacturing Approach at Nabil Factory

Production planning techniques at Nabil Factory play a critical role not only in determining how much to produce, but also in identifying **which manufacturing approach best fits the operational and market realities** of the Chicken Burger product. Through systematic analysis of demand, capacity, inventory, and scheduling, the planning system clearly supports the adoption of a **hybrid Make-to-Order (MTO) supported by Make-to-Stock (MTS)** strategy.

Role of Demand Planning and Forecasting

Demand planning at Nabil Factory is primarily based on **stable and growing monthly contracts**, especially with restaurant customers. This creates a predictable demand pattern, but one that is also **continuously increasing**. Planning techniques reveal that while demand is forecastable, it is also **time-critical**, as restaurant customers require **immediate delivery with zero tolerance for delays**.

From a planning perspective, this combination of predictable volume and strict delivery requirements makes a pure Make-to-Order system risky. Although orders are known in advance, production lead times—driven by batch processing and freezing—are too long to rely solely on reactive production. As a result, planning techniques highlight the need for **pre-planned production and buffering**, leading to the use of MTS elements alongside MTO.

Customer Requirements and Product Customization

Planning analysis also shows that restaurant customers require **high levels of customization**, including differences in weight, thickness, seasoning, and whether the product is cooked or uncooked. These variations are operationally significant and directly affect routing and processing decisions.

Through Master Production Scheduling (MPS) and routing analysis, planners can clearly see that these differences prevent full product standardization. This insight rules out a pure Make-to-Stock approach, as producing standardized inventory in advance would increase the risk of non-conforming products. Therefore, planning techniques support the use of **Make-to-Order as the dominant manufacturing approach** for restaurant customers.

Capacity Planning and Line Structure

Capacity planning reveals that production at Nabil Factory is constrained not by demand uncertainty, but by **system structure**. Multiple products share the same production lines, and although changeovers between molds are relatively quick, the system remains **batch-based**. This means production must be carefully planned to avoid congestion and imbalance across lines.

Planning techniques show that without structured planning, simultaneous orders across multiple products would overwhelm shared resources. As a result, production cannot be triggered purely by customer orders in real time. Instead, it must be coordinated through aggregate planning and scheduling. This insight further confirms that **pure MTO is operationally impractical**, reinforcing the need for a hybrid approach.

Inventory Planning and Supplier Lead Times

Inventory analysis plays a decisive role in identifying the appropriate manufacturing approach. The two-month lead time for chicken meat imported from Brazil introduces significant supply risk. Planning techniques clearly show that reducing inventory under these conditions would expose the system to severe production interruptions.

As a result, inventory buffers such as the 700-ton safety stock are not optional but essential. This requirement directly supports the inclusion of **Make-to-Stock logic** within the overall MTO framework. In other words, planning techniques demonstrate that inventory is not a sign of inefficiency, but a **strategic necessity** given supplier constraints

Scheduling Reality and Flow Characteristics

Integration with Operations Management Principles

Recommended Production Planning Technique and Production Approach

Based on my analysis the most suitable production approach for the Chicken Burger line at Nabil Factory is a **Make-to-Order (MTO) approach supported by strategic inventory buffers** rather than pure Make-to-Stock or Engineer-to-Order.

This recommendation is driven by the nature of demand from restaurant customers Orders are placed on a **monthly contractual basis** with clearly defined minimum quantities and **strict delivery requirements** Each restaurant specifies customized product attributes such as weight, seasoning, and cooking condition, which makes pre-producing standardized inventory inefficient and risky. As a result, production is triggered **only after order confirmation**, aligning closely with MTO logic.

At the same time, the system does not operate as a pure MTO environment. Due to long supplier lead times for frozen chicken and the critical importance of delivery reliability, **raw material inventory buffers are intentionally maintained**. This hybrid structure allows Nabil Factory to respond quickly once orders are confirmed, without exposing the system to material shortages. Therefore, the production approach can be described as **order-driven production with supply-side buffering**, which is fully consistent with the operational constraints of the factory.

From a critical perspective, this approach is appropriate because delivery delays are not acceptable under restaurant contracts, and customization is non-negotiable. A pure Make-to-Stock approach would increase the risk of obsolescence and waste, while Engineer-to-Order is unnecessary given the standardized production processes.

Recommended Production Planning Technique

The most effective production planning technique for this case is a **medium-term, forecast-based planning system integrated with capacity-constrained execution and push-controlled release**, supported by detailed scheduling and MRP.

At the strategic and tactical level, **monthly forecasting** remains essential. Even though production is order-driven, forecasting plays a critical role in:

- Capacity planning
- Workforce planning
- Raw material procurement
- Safety stock determination

Forecasting in this context is not used to produce inventory blindly, but to **prepare the system** for expected demand volumes. Given the relatively stable but growing demand from restaurant customers, this approach ensures that the freezer capacity and supporting resources are not overloaded.

At the operational level, production planning must explicitly recognize the **freezer as the system bottleneck**, as identified through the Value Stream Map and TOC analysis. Therefore, planning decisions are capacity-driven rather than volume-driven. Production orders are released using a **push-based logic upstream of the freezer**, where material is scheduled according to the plan rather than pulled by downstream consumption. This is necessary due to batch freezing, food safety constraints, and long processing times that prevent real-time pull signaling.

However, this push logic is not uncontrolled. It is regulated by:

- Capacity checks at the bottleneck
- FIFO sequencing rules
- Time-based release buffers
- Overtime as a controlled flexibility mechanism

This results in a **disciplined push system**, rather than an uncontrolled push environment.

Justification of Push-Oriented Planning with Capacity Control

Although pull systems are often preferred in lean environments, a **pure pull system is not suitable** for the Chicken Burger production line. The presence of batch processing, long supplier lead times, and strict delivery commitments makes pull-based replenishment impractical and risky. Instead, the system relies on **planned release of work**, supported by inventory buffers, to ensure stability and continuity.

From a systems engineering perspective, the selected planning technique prioritizes:

- Delivery reliability over minimum inventory
- System stability over maximum responsiveness

- Throughput protection over local efficiency

This trade-off is justified by the operational context and contractual obligations of Nabil Factory.

Critical Evaluation of the Recommendation

From a critical standpoint, the recommended MTO-based, push-controlled planning system is **effective but not without limitations**. While it ensures high delivery reliability and supports customization, it also leads to:

- High inventory levels
- Long lead times
- Dependence on overtime to absorb variability

However, given the system constraints and food industry requirements, these limitations represent **managed trade-offs rather than planning failures**. The advanced planning improvements further mitigate these weaknesses without undermining the strengths of the current system.

Final Recommendation

In conclusion, the most suitable production planning approach for the Chicken Burger production line at Nabil Factory is an **order-driven (MTO) production strategy supported by forecast-based planning, capacity-constrained scheduling, and controlled push execution**. This combination balances customization, delivery reliability, and system stability, making it both operationally realistic and strategically sound within the constraints of high-volume food manufacturing.

Use of Value Stream Mapping (VSM) as a Production Plan and Compare the Recommended Approach with the Current VSM

Use of Value Stream Mapping (VSM) as a Production Planning Tool

is traditionally used as a lean analysis tool to visualize material and information flows, identify waste, and distinguish between value-adding and non-value-adding activities. However, in the context of the Chicken Burger production line at **Nabil Factory**, VSM also functions as a **practical production planning support tool**, rather than only a diagnostic map.

In this project, VSM was used to:

- Visualize the **end-to-end production flow**, from raw material receiving to dispatch.
- Identify **capacity constraints**, particularly the freezer as the system bottleneck.
- Quantify **processing time versus waiting time**, revealing that only a very small portion of total lead time is value-adding.
- Support planning decisions related to **batch sizing, scheduling logic, and resource allocation**.

operational data cycle times, batch sizes and WIP levels and lead times with planning decisions, the VSM effectively bridges **operations execution and production planning**. This aligns with manufacturing systems engineering principles, where planning must be grounded in the physical realities of the production system rather than abstract forecasts alone.

Current-State VSM as a Reflection of the Existing Production Plan

The **current-state VSM** represents how production planning is actually executed at Nabil Factory.

Key characteristics of the current-state VSM include:

- **Push-based production control**
  Production is released based on monthly plans and forecasts rather than real-time downstream demand signals.
- **Batch-oriented flow**
  Several processes, especially freezing, operate in large batches. This results in significant waiting and WIP accumulation upstream of the freezer.

- **Freezer as the dominant bottleneck**
  With a fixed freezing time of 60 minutes per batch, the freezer dictates system throughput and pacing.
- **Very low activity ratio (~0.12%)**
  The vast majority of lead time is consumed by waiting and storage rather than processing.

From a production planning perspective, the current VSM shows a system that is **stable and reliable**, but also **inefficient in terms of flow**. Planning effectiveness is achieved through inventory, overtime, and batching rather than through synchronized flow.

Recommended Approach: VSM as a Future-Oriented Planning Tool

In the recommended approach, VSM is not treated only as a snapshot of current performance, but as a **design tool for improving production planning decisions**.

The **future-state VSM** translates strategic planning objectives into operational structure by addressing the main weaknesses identified in the current state.

Key planning-oriented improvements include:

1. Lead Time Reduction through Supplier Strategy

- Supplier lead time is reduced from approximately **two months to two weeks** by shifting to regional suppliers.
- This directly reduces raw material inventory and total system lead time.

2. Improved Internal Flow through Layout Integration

- Mixing and forming processes are physically and logically integrated.
- Manual transport and intermediate WIP are reduced.
- Production planning benefits from **more predictable internal flow** and reduced variability.

3. Bottleneck-Aware Planning (TOC Alignment)

- The freezer remains the bottleneck, but upstream flow is better synchronized.
- Instead of overfeeding the bottleneck with excess WIP, controlled release improves flow efficiency.
- Planning shifts from "maximize utilization everywhere" to **"protect and pace the constraint."**

4. Reduced Non-Value-Added Time

- Although processing time remains largely unchanged (due to food safety constraints), non-value-added time is significantly reduced.
- Activity ratio improves from **0.12% to approximately 0.46%**, representing a major improvement in flow efficiency.

Comparison: Current VSM vs Recommended VSM

Aspect	Current-State VSM	Recommended VSM
Planning logic	Push-based, forecast-driven	Controlled push, constraint-driven
Bottleneck management	Protected by excess WIP	Protected by synchronized release
Inventory levels	Very high (RM, WIP, FG)	Reduced, better positioned
Lead time	Extremely long (weeks/months)	Significantly reduced
Activity ratio	~0.12%	~0.46%
Flow efficiency	Very low	Improved but realistic
Planning role of VSM	Descriptive	Prescriptive and decision-oriented

Critical Evaluation

The current VSM reflects a production planning system that **prioritizes delivery reliability over flow efficiency**, which is understandable given:

- Strict restaurant delivery commitments
- Long supplier lead times
- Batch freezing and food safety constraints

However, the recommended VSM demonstrates that **planning effectiveness can be improved without increasing risk**. By using VSM as a production planning guide, the system shifts from compensating for inefficiencies (inventory, overtime) toward **structurally reducing their causes**.

Importantly, the recommended approach does not propose unrealistic changes such as eliminating batch freezing or implementing a pure pull system. Instead, it uses VSM to align:

- Capacity planning
- Scheduling
- Inventory positioning
- Bottleneck control

In this project, Value Stream Mapping proves to be more than a lean visualization tool. It acts as a **core production planning instrument**, enabling informed decisions about capacity inventory, scheduling, and flow control.

The comparison between the current and recommended VSM shows that while Nabil Factory's existing planning system is reliable and robust, it is not fully optimized. The recommended VSM demonstrates how planning can evolve from **buffer-based stability** toward **flow-based efficiency**, while preserving food safety, delivery reliability, and operational realism

Task VI

Production Planning Engineer

Role Description

The Production Planning Engineer is responsible for translating demand forecasts and customer contracts into **feasible production plans**.

Contribution to the Chicken Burger Line

the Production Planning Engineer:

- Converts monthly restaurant contracts into master production schedules (MPS)
- Balances make-to-order restaurants and make-to-stock retail requirements
- Aligns production volumes with freezer capacity, which is the system bottleneck

This role directly supports the effectiveness of forecasting accuracy, throughput stability, and delivery reliability

Industrial / Manufacturing Systems Engineer

Role Description

The Industrial or Manufacturing Systems Engineer focuses on process flow, system efficiency, and bottleneck management. This role applies systems engineering principles such as TOC throughput analysis, cycle time, and lead time reduction.

Contribution to the Chicken Burger Line

In the Chicken Burger line, this engineer:

- Identifies the **freezer as the system constraint**
- Analyzes **WIP accumulation and batch behavior**
- Evaluates trade-offs between **throughput, inventory, and lead time**

This role ensures that production planning decisions are **system-optimized**, not just locally optimized

Quality / Six Sigma Engineer

Role Description

The Quality or Six Sigma Engineer is responsible for **process stability, defect reduction, and quality consistency**, using data-driven improvement methods.

Contribution to the Chicken Burger Line

For the Chicken Burger process, this role:

- Applies **DMAIC** to reduce defects at frying and cooking stages
- Stabilizes process variability that affects scheduling and delivery
- Reduces scrap and rework, which directly improves **OEE and effective capacity**

By reducing variability, this role strengthens the effectiveness of **capacity planning, scheduling accuracy, and inventory control**

5. Supply Chain / Materials Engineer

Role Description

The Supply Chain or Materials Engineer manages **raw material availability, safety stock, and supplier coordination**, which is essential in long lead-time environments.

Contribution to the Chicken Burger Line

In this case, the engineer:

- Manages 700-ton safety stock for frozen chicken
- Aligns purchasing cycles with P-system periodic review logic
- Coordinates long supplier lead times with production plans

This role is critical to maintaining production continuity in a system that relies on push-based planning and batch processing

ERP / Odoo System Engineer

Role Description

The ERP or Systems Engineer ensures proper **configuration, integration, and use of Odoo** to support planning and execution.

Contribution to the Chicken Burger Line

This engineer:

- Maintains accurate **BOMs and routings**
- Ensures MRP calculations reflect real capacities and lead times
- Integrates forecasting, inventory, and manufacturing orders

Without this role, the planning methods described could not be executed consistently or reliably.

Integrated Impact on Manufacturing System Success

The success of the Chicken Burger manufacturing system depends not on one role, but on the **coordination between engineering roles**:

- Production Planning ensures feasibility
- Systems Engineering ensures flow optimization
- Quality Engineering ensures stability
- Maintenance ensures availability
- Supply Chain ensures material readiness
- ERP Engineering ensures system integration

Task VII

Introduction

The purpose is not to redesign the production system from scratch, but to enhance the effectiveness of planning and control decisions by addressing the structural inefficiencies identified earlier namely long lead times high inventory levels, reliance on overtime, and batch based flow while preserving the system's key strengths: delivery reliability, food safety, and production continuity.

The proposed advanced planning framework focuses on six interrelated dimensions:

- Scheduling
- Capacity planning
- Material Requirements Planning (MRP)
- Inventory management
- Forecasting
- System-level integration

Together, these elements form a coherent, constraint-aware planning system aligned with the operational realities of high-volume food manufacturing.

1. Advanced Scheduling Plan

Current Scheduling Limitations

The current scheduling approach at Nabil Factory is primarily weekly and batch-oriented, supported by push-based release of work into the system. While this approach ensures that production targets are met, it also creates several systemic inefficiencies:

- Excessive work-in-process (WIP) accumulation upstream of the freezer
- Long internal waiting times
- Limited visibility of true system capacity
- Heavy dependence on overtime to absorb variability

The Value Stream Map developed in clearly shows that scheduling decisions are not synchronized with the system constraint, resulting in high lead times despite relatively short processing times.

Proposed Scheduling Strategy: Constraint-Driven Scheduling

The advanced scheduling plan is based on explicit synchronization with the system bottleneck, which in this case is the freezer. According to manufacturing systems engineering principles, overall system performance is governed by the slowest or most constrained resource; therefore, scheduling decisions must be structured around this constraint.

Under the proposed approach:

- The freezer becomes the pacemaker process for the entire production line
- Work is released into upstream processes only at a rate that the freezer can absorb
- Scheduling focuses on timing of release, not increasing batch size or speed

This approach does not attempt to eliminate batch processing, which is technically required for freezing and food safety. Instead, it controls flow through disciplined release rules.

Scheduling Rules and Execution Logic

To ensure operational feasibility and stability:

- Weekly schedules remain fixed to maintain execution discipline
- FIFO sequencing is maintained due to food safety and traceability requirements
- Release buffers are defined in time units, not volume units
- Overtime is triggered only when freezer utilization exceeds predefined thresholds

Expected Scheduling Improvements

Performance Dimension	Current State	Advanced Scheduling
WIP before freezer	High	Reduced by 20–30%
Internal waiting time	High	Reduced by 15–20%
Schedule stability	Medium	High
Throughput reliability	High	Maintained

This approach improves flow efficiency without sacrificing delivery reliability, which is critical in a contract-driven food manufacturing environment.

2. Advanced Capacity Planning

Current Capacity Planning Assessment

Capacity planning at Nabil Factory is adequate but tight, as demonstrated inInstalled capacity—particularly at the freezer—is sufficient to meet average demand, yet the system relies heavily on overtime as a flexibility mechanism. This indicates that capacity is managed reactively rather than strategically.

Key limitations include:

- Lack of explicit capacity cushion management
- Limited forward-looking load analysis
- High sensitivity to demand fluctuations

Proposed Capacity Planning Framework

The advanced capacity planning approach introduces explicit management of capacity cushions, transforming capacity from a reactive constraint into a controlled planning variable.

Key elements include:

- Maintaining a planned 15% capacity cushion, aligned with observed utilization levels
- Monitoring freezer load on a weekly basis
- Allowing overtime only when utilization exceeds 90%
- Differentiating between structural capacity needs and short-term variability

This approach recognizes that capacity should absorb variability, not mask inefficiencies.

Quantified Capacity Planning Improvements

Metric	**Current**	**Advanced Plan**
Average utilization	**~85%**	**Controlled at 80–85%**
Overtime usage	**Daily**	**Reduced by 20–25%**

Metric	Current	Advanced Plan
Capacity predictability	Medium	High
Cost volatility	High	Reduced

This improves both economic performance and planning reliability.

3. Advanced Material Requirements Planning (MRP)

Current MRP Performance

MRP at Nabil Factory is well implemented through the Odoo ERP system, with accurate BOMs and material explosion logic. However, its effectiveness is constrained by:

- Long supplier lead times (approximately two months for chicken)
- Rigid monthly planning cycles
- Limited responsiveness to external disruptions

As a result, MRP decisions often prioritize availability over optimization, contributing to high inventory levels.

Proposed MRP Enhancement: Rolling MRP with Frozen Horizon

The advanced MRP plan introduces a rolling planning mechanism that balances stability and responsiveness.

Key improvements:

- Monthly planning horizon remains unchanged
- Weekly rolling updates are introduced
- The first two weeks are frozen to protect execution
- Weeks three to six are dynamically adjusted based on updated forecasts

Expected MRP Performance Improvements

METRIC	CURRENT	ADVANCED MRP
MATERIAL SHORTAGES	Rare	Eliminated

OVER-PURCHASING	**Medium**	**Reduced by 10–15%**
PLANNING RESPONSIVENESS	**Medium**	**High**
EXECUTION STABILITY	**High**	**Maintained**

This approach improves decision quality without increasing system nervousness.

4. Advanced Inventory Management Strategy

Current Inventory Assessment

Inventory at Nabil Factory is used as a strategic buffer, particularly due to long supplier lead times and strict delivery commitments. While effective in protecting service levels, inventory contributes significantly to:

- Long lead times
- High capital tie-up
- Reduced flow efficiency

inventory is compensating for structural constraints rather than supporting lean flow.

Proposed Inventory Strategy: Strategic Buffer Rebalancing

Rather than reducing inventory indiscriminately, the advanced plan focuses on buffer positioning and balance.

Key actions:

- Maintain raw material safety stock at current levels to manage supplier risk
- Reduce WIP buffers upstream of the freezer
- Shift buffering downstream where demand visibility is higher
- Use lead-time reduction as the primary lever for inventory reduction

According to Little's Law, reducing lead time directly reduces inventory without compromising throughput.

Inventory Impact Summary

Inventory Category	Expected Change
Raw material	**Stable**
WIP	**Reduced by ~20%**

Finished goods	**Slight reduction**
Total inventory	**Reduced by 10–15%**

This improves flow while maintaining delivery reliability.

5. Advanced Forecasting Integration

Current Forecasting Limitations

Forecasting at Nabil Factory is primarily monthly and historically accurate under stable conditions. However, it is vulnerable to:

- External shocks market disruptions
- Sudden demand changes
- Behavioral overreactions leading to overstock

Proposed Forecasting Framework

The advanced forecasting approach shifts focus from prediction accuracy to decision support.

Key enhancements:

- Monthly statistical forecast remains the baseline
- Event-based overrides are introduced
- Demand scenarios are defined (normal, peak, disruption)
- Forecast outputs are directly linked to capacity and inventory decisions

Forecasting Performance Improvements

Metric	Current	Advanced Forecasting
Sensitivity to shocks	High	Reduced
Overstock risk	Medium	Lower
Planning confidence	Medium	High

This approach improves robustness rather than chasing perfect accuracy.

6. Integrated System-Level Impact

The proposed advanced planning and control system:

- Does not eliminate batch processing
- Does not compromise food safety
- Does not require capital investment
- Preserves delivery reliability

Instead, it:

- Improves flow coordination
- Reduces unnecessary buffers
- Enhances decision transparency
- Strengthens system resilience

Overall Performance Improvement Summary

Dimension	***Expected Improvement***
Delivery reliability	**Maintained**
Lead time	**Reduced by 15–25%**
Inventory	**Reduced by 10–15%**
Overtime	**Reduced by 20–25%**
Flow efficiency	**Improved**
System robustness	**Increased**

Resource Allocation Strategy

Current Resource Allocation Assessment

Resource allocation at Nabil Factory is currently functional but highly reactive. Labor and equipment are generally sufficient to meet production demand; however, allocation decisions are primarily driven by short-term scheduling pressure rather than system-wide optimization. The analysis conducted using chase, level, and mixed workforce strategies showed that while demand variability exists, the production system relies heavily on overtime and informal adjustments to absorb fluctuations.

Key limitations of the current resource allocation approach include:

- Frequent reliance on overtime as the primary flexibility mechanism
- Limited alignment between workforce allocation and system constraints
- Reactive reassignment of labor to address bottlenecks rather than preventing them
- Workforce instability under chase-like behavior, increasing cost and operational risk

Although these practices ensure delivery reliability, they increase operating costs and reduce workforce efficiency.

Proposed Resource Allocation Strategy: Level-Based, Constraint-Focused Allocation

The advanced resource allocation strategy is built on a **level workforce approach**, supported by selective overtime and constraint-aware task assignment. This approach was selected based on quantitative analysis, which showed that the level strategy resulted in the **lowest total cost** while maintaining operational stability.

Key Principles of the Proposed Strategy

- Maintain a **stable core workforce** across planning periods
- Allocate labor based on **system constraints**, not local utilization
- Use overtime only as a **controlled buffer**, not a default solution
- Align labor allocation directly with scheduling and capacity plans

Workforce Allocation Logic

Based on the level strategy analysis:

- Workforce size is fixed at **26 employees**
- No hiring or layoffs are planned during normal operations
- Demand variability is absorbed through:
    - Limited overtime
    - Minor undertime during low-load periods
- Cross-trained operators allow flexible reassignment across processes

This approach preserves workforce skill, reduces training and turnover costs, and improves execution consistency.

Constraint-Based Resource Assignment

In alignment with the Theory of Constraints:

- The freezer, as the system bottleneck, receives **priority in labor allocation**
- Upstream resources are scheduled to support, not overload, the freezer
- Labor is reassigned proactively to:
    - Prevent upstream WIP accumulation
    - Ensure continuous freezer utilization
- Resource conflicts are resolved at the system level rather than locally

This prevents the common inefficiency of maximizing local utilization at the expense of overall flow.

Quantified Resource Allocation Improvements

Metric	Current State	Advanced Resource Allocation
Workforce stability	Medium	High
Overtime usage	Daily	Reduced by 20–25%
Labor cost variability	High	Reduced
Bottleneck support	Reactive	Proactive
Workforce utilization balance	Uneven	Optimized

Resource Allocation Impact Summary

The advanced resource allocation strategy:

- Reduces labor cost volatility
- Improves workforce morale and skill retention
- Enhances alignment between labor, scheduling, and capacity
- Supports long-term operational sustainability

Rather than reacting to demand variability through constant workforce changes, the system absorbs variability through disciplined planning and controlled flexibility.

Critical thinking

The advanced planning and control framework developed in this task transforms the Chicken Burger production system at Nabil Factory from a compensatory system one that relies on inventory and overtime to absorb inefficiencies into a managed and coordinated system

By aligning scheduling,capacity planning, MRP inventory management, and forecasting around system constraints and decision integration, the factory can sustain its strong delivery performance while significantly improving efficiency, cost control, and long-term operational sustainabilityThis represents a mature application of manufacturing systems engineering principles to a real-world high-volume food production environment.

Task VIII

Use of Software

This project made use of several industrial engineering and operations management software tools to support production planning, analysis, visualization, and decision-making. Each tool was selected based on its relevance to the planning problem and its ability to model real manufacturing system behavior.

8.1 Microsoft Project – Scheduling and Resource Assignment

Microsoft Project was used to develop and manage the detailed production schedule for the Chicken Burger production line. All major processing stages (Microwave, Mixing, Forming, Coating, Frying, Cooking, Freezing, and Packing) were modeled as tasks, with three manufacturing orders (A, B, and C) flowing through the same shared resources.

Task durations, precedence relationships, and resource assignments were defined to reflect the real production sequence and machine-sharing constraints. This allowed visualization of resource conflicts, validation of scheduling feasibility, and confirmation of the freezer as the system bottleneck.

		Name	Duration	Number1	Start	Finish	Predecessors	Task Calendar	Resource Names
1		Microwave	0.114 days	0.0	1/1/26 8:00 AM	1/1/26 10:30 AM		Nabil	
2		JOB A	0.027 days	10.0	1/1/26 8:00 AM	1/1/26 8:35 AM			microwave_op1;microwave...
3		JOB B	0.017 days	5.0	1/1/26 8:00 AM	1/1/26 8:22 AM			microwave_op2;microwave...
4		JOB C	0.114 days	7.0	1/1/26 8:00 AM	1/1/26 10:30 AM			microwave_op1;microwave...
5		Mixing	0.286 days	0.0	1/1/26 8:22 AM	1/1/26 3:40 PM		Nabil	
6		A	0.064 days	10.0	1/1/26 8:35 AM	1/1/26 10:00 AM	2		mixing_op1;mixing_op2
7		C	0.036 days	5.0	1/1/26 8:22 AM	1/1/26 9:10 AM	3		mixing_op1;mixing_op2
8		C	0.189 days	7.0	1/1/26 10:30 AM	1/1/26 3:40 PM	4		mixing_op1;mixing_op2
9		Forming	0.377 days	0.0	1/1/26 9:10 AM	1/2/26 9:27 AM		Nabil	
10		A	0.339 days	10.0	1/1/26 10:00 AM	1/2/26 9:27 AM	6		forming_op
11		B	0.177 days	5.0	1/1/26 9:10 AM	1/1/26 2:03 PM	7		forming_op
12		C	0.038 days	7.0	1/1/26 3:40 PM	1/1/26 4:30 PM	8		forming_op
13		Coating I	4.634 days?	0.0	1/1/26 2:03 PM	1/14/26 11:00 AM		Nabil	
14		A	0.88 days	10.0	1/2/26 9:27 AM	1/6/26 1:49 PM	10		Coating I _op
15		B	0.198 days	5.0	1/1/26 2:03 PM	1/2/26 9:24 AM	11		Coating I _op
16		C	0.364 days?	7.0	1/1/26 4:30 PM	1/2/26 4:30 PM	12		Coating I _op
17		Coating II	1.999 days?	0.0	1/2/26 9:24 AM	1/7/26 2:23 PM		Nabil	
18		JOB A	0.389 days	10.0	1/6/26 1:49 PM	1/7/26 2:23 PM	14		Coating II_op
19		JOB B	0.202 days	5.0	1/2/26 9:24 AM	1/2/26 2:51 PM	15		Coating II_op
20		JOB C	0.364 days?	7.0	1/2/26 4:30 PM	1/5/26 4:30 PM	16		Coating II_op
21		Frying	2.261 days?	0.0	1/2/26 2:51 PM	1/8/26 4:35 PM		Nabil	
22		A	0.464 days	10.0	1/7/26 2:23 PM	1/8/26 4:35 PM	18		Frying_op
23		B	0.236 days	5.0	1/2/26 2:51 PM	1/5/26 11:03 AM	19		Frying_op
24		C	0.364 days?	7.0	1/5/26 4:30 PM	1/6/26 4:30 PM	20		Frying_op
25		Cooking	2.786 days?	0.0	1/5/26 11:03 AM	1/13/26 8:20 AM		Nabil	
26		A	0.761 days	10.0	1/8/26 4:35 PM	1/13/26 8:20 AM	22		Cooking_op
27		B	0.383 days	5.0	1/5/26 11:03 AM	1/6/26 11:28 AM	23		Cooking_op
28		C	0.364 days?	7.0	1/6/26 4:30 PM	1/7/26 4:30 PM	24		Cooking_op
29		Freezing	2.888 days?	0.0	1/6/26 11:28 AM	1/14/26 11:00 AM		Nabil	
30		A	0.485 days	10.0	1/13/26 8:20 AM	1/14/26 11:00 AM	26		Freezing_op
31		B	0.258 days	5.0	1/6/26 11:28 AM	1/7/26 9:08 AM	27		Freezing_op
32		C	0.364 days?	7.0	1/7/26 4:30 PM	1/8/26 4:30 PM	28		Freezing_op

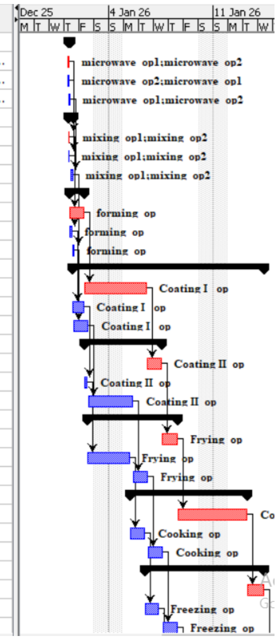


Figure 20 Gantt chart & task sequencing

	Name	RBS	Type
	microwave_op1		Work
	microwave_op2		Work
	mixing_op1		Work
	mixing_op2		Work
	forming_op		Work
	Coating I _op		Work
	Coating II_op		Work
	Frying_op		Work
	Cooking_op		Work
	Packing_op1		Work
	Packing_op2		Work
	Packing_op3		Work
	Freezing_op		Work

Figure 21 resource usage

8.2 Odoo ERP – MRP, BOM, and Manufacturing Orders

Odoo ERP was used to model the production structure and material planning logic of the factory. Products and raw materials were created, and a detailed Bill of Materials (BOM) was defined for the Chicken Burger product. Routing and work centers were configured to represent the actual production processes.

Manufacturing Orders (MO) were generated based on demand, enabling material explosion, inventory reservation, and execution tracking. This supported analysis of MRP effectiveness, inventory availability, and push-based production behavior.

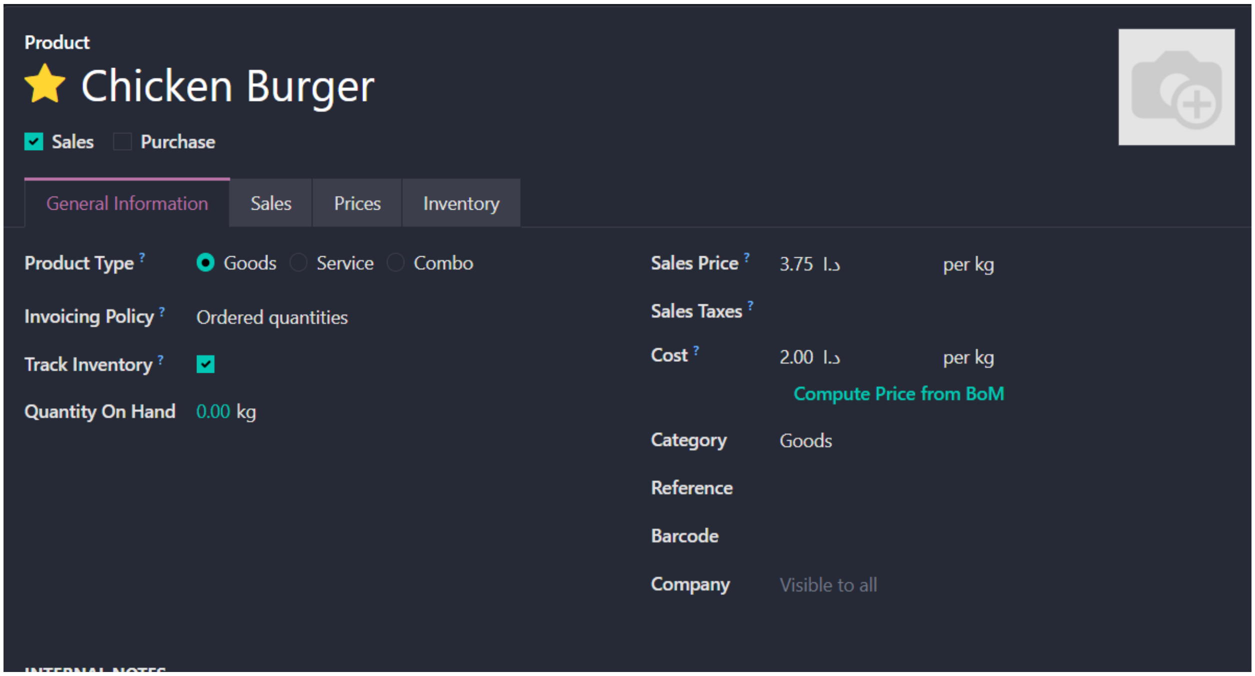


Figure 22 Chicken Burger

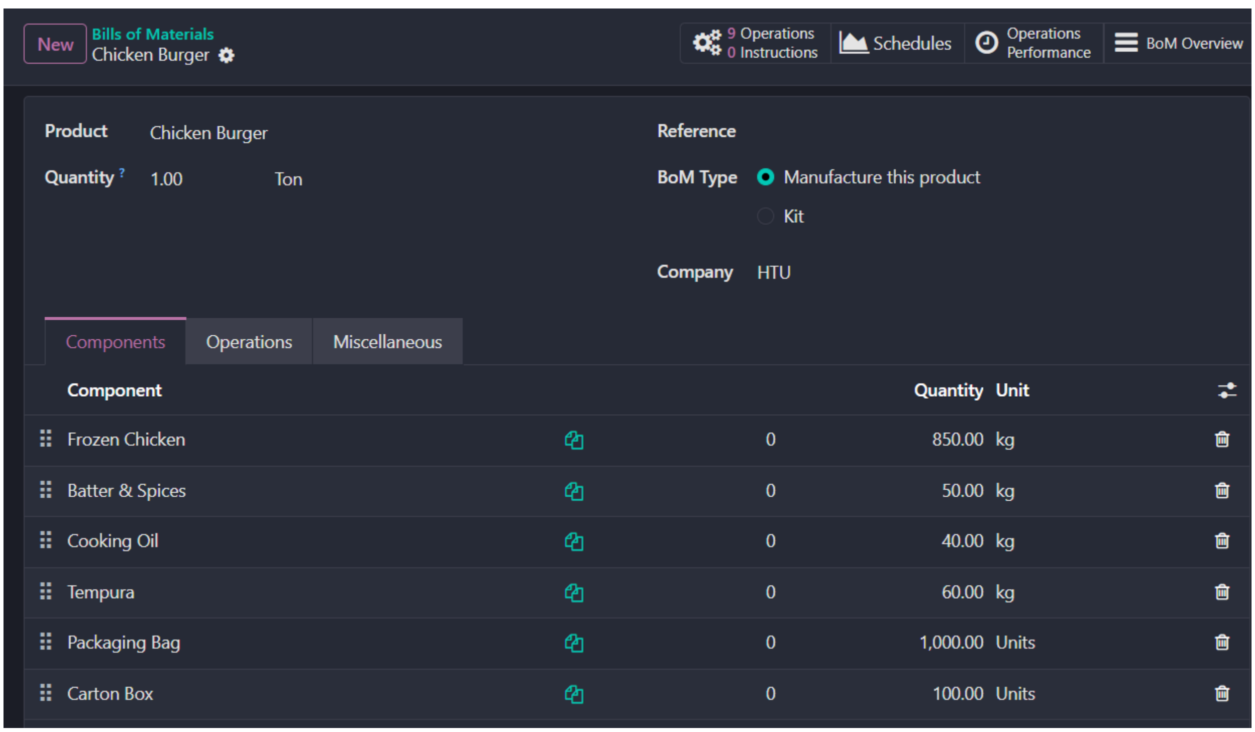
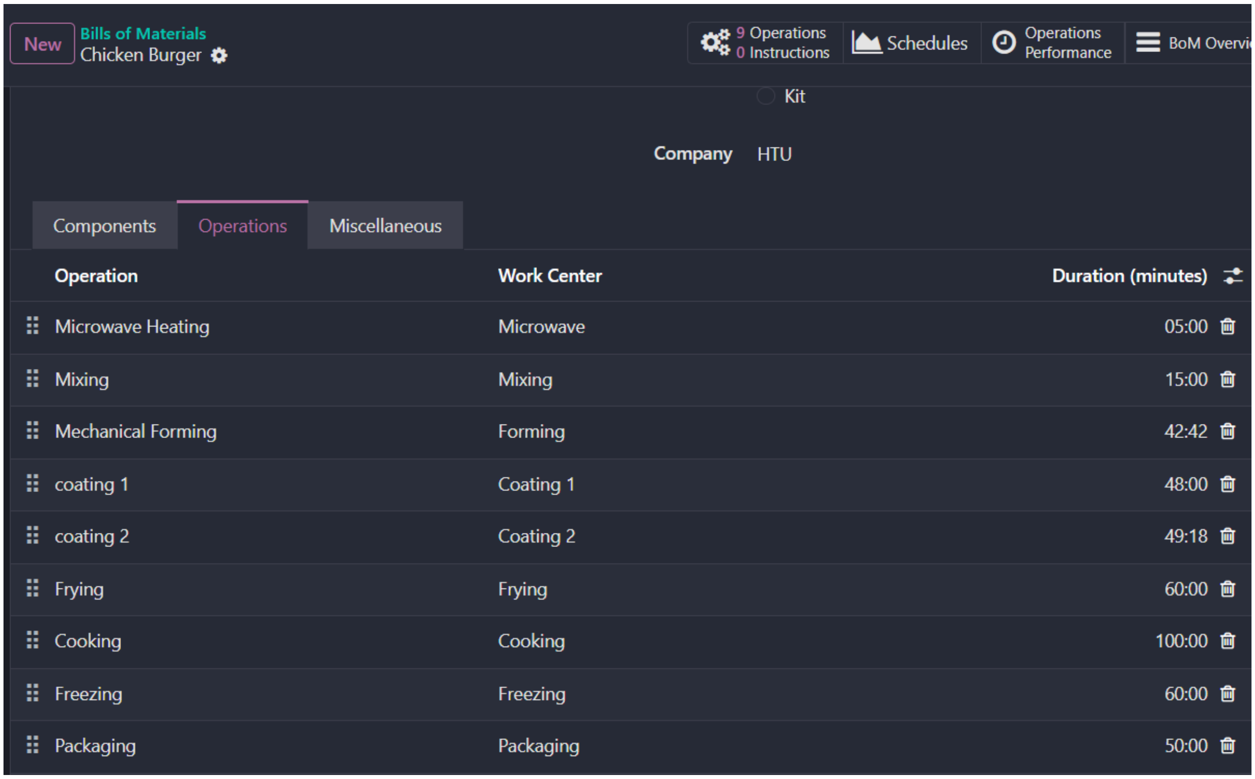


Figure 23 Bill of Materials (BOM)



Operation	Work Center	Duration (minutes)
Microwave Heating	Microwave	05:00
Mixing	Mixing	15:00
Mechanical Forming	Forming	42:42
coating 1	Coating 1	48:00
coating 2	Coating 2	49:18
Frying	Frying	60:00
Cooking	Cooking	100:00
Freezing	Freezing	60:00
Packaging	Packaging	50:00

Figure 24 Routing and work centers.

8.3 Minitab – Demand Forecasting

Minitab was used to perform demand forecasting based on historical sales data. Time-series analysis was applied to estimate monthly demand and identify demand stability and seasonality. The resulting forecast provided the baseline input for capacity planning, MRP, and inventory decisions.

Month	Demand	AVER1	RESI1	SMOO2	RESI2	FORE2	RESI3	FITS3
1	387.743	*	*	362.581	31.4523	353.906	16.4381	371.305
2	450.000	*	*	380.065	87.4187	353.906	81.3971	368.603
3	330.272	389.338	*	370.106	-49.7933	353.906	-35.6289	365.901
4	330.152	370.141	-59.1863	362.115	-39.9544	353.906	-33.0464	363.198
5	350.103	336.842	-20.0381	359.713	-12.0123	353.906	-10.3930	360.496
6	289.475	323.243	-47.3671	345.665	-70.2379	353.906	-68.3188	357.794
7	351.405	330.328	28.1620	346.813	5.7399	353.906	-3.6863	355.092
8	381.523	340.801	51.1948	353.755	34.7093	353.906	29.1333	352.389
9	334.485	355.804	-6.3164	349.901	-19.2706	353.906	-15.2026	349.687
10	334.013	350.007	-21.7915	346.723	-15.8884	353.906	-12.9723	346.985
11	390.639	353.045	40.6321	355.507	43.9153	353.906	46.3560	344.283
12	347.504	357.385	-5.5411	353.906	-8.0023	353.906	5.9238	341.581

Figure 25 Demand Forecasting

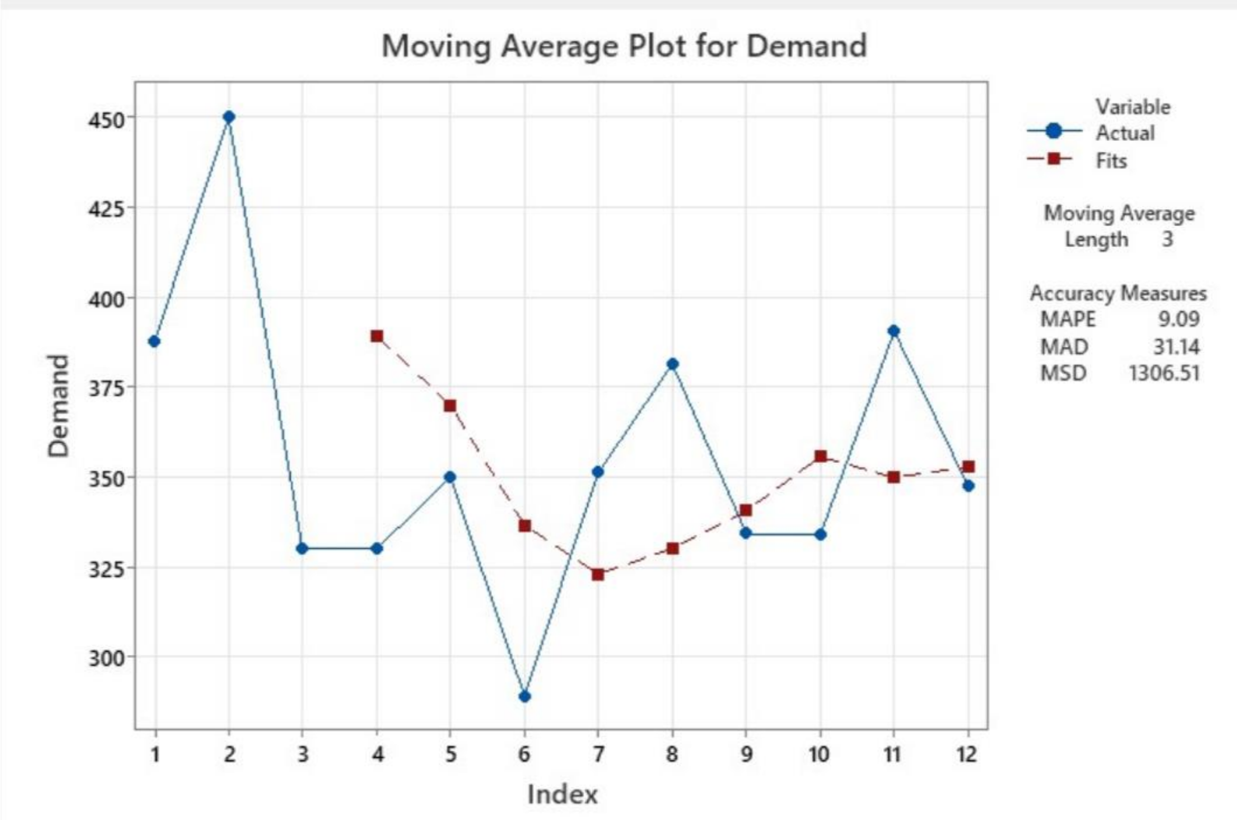


Figure 26 series plot and demand

8.4 Microsoft Excel – Value Stream Mapping and Strategy Comparison

Microsoft Excel was used for analytical modeling and visualization. A Value Stream Map (VSM) was developed to quantify process times, waiting times, inventory accumulation, and total lead time. This analysis revealed significant flow inefficiencies and supported bottleneck-focused improvements.

Excel was also used to compare Chase, Level, and Mixed production strategies. Workforce levels, hiring and layoff costs, overtime, and total cost were calculated, supporting the selection of the Level strategy as the most stable and cost-effective approach.

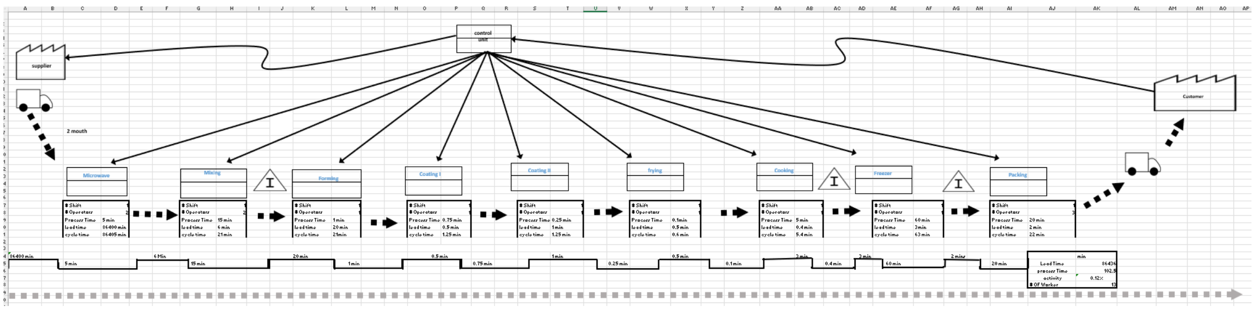


Figure 27 Value Stream Map (VSM) of the Chicken Burger production line.

Period	data	1	2	3	4	Total	
Demand (hours)		10,000	7,000	8,500	7,600		
Employees Needed		35	24	30	27		
Workforce Level	26	35	24	30	27		
Hires		9	0	6	0	15	IF(C4>B4,(C4-B4),"0")
Layoffs		0	11	0	3	14	IF(C4<B4,(B4-C4),"0")
Regular Cost ($)	800	28,000	19,200	24,000	21,600		
Hiring Cost ($)	900	8,100	-	5,400	-		
Layoff Cost ($)	1,200	-	13,200	-	3,600		
Total Cost ($)		**36,100**	**32,400**	**29,400**	**25,200**	**123,100**	

Figure 28 Chase Strategy

Period	data	1	2	3	4	Total
Demand (hours)		10,000	7,000	8,500	7,600	
Demand (Employees)	26	35	24	30	27	
Workforce Level	700	26	26	26	26	14
Undertime (Employees)		0	2	0	0	IF(C4>C3,C4-C3,"0")
Overtime (Employees)		9	0	4	1	IF(C4<C3,C3-C4,"0")
Regular Cost ($)	800	28,000	19,200	20,800	21,600	
Undertime Cost ($)	8	-	1,600	-	-	
Overtime Cost ($)	15	3,510	-	1,560	390	104
Hiring Cost ($)		24,000	-	-	-	
Total Cost ($)		55,510	20,800	22,360	21,990	120,660

Figure 29 Level

Period	data	1	2	3	4	Total
Demand (hours)		10,000	7,000	8,500	7,600	
Demand (Employees)		35	24	30	27	
Workforce Level	26	35	35	43	30	
Undertime (Employees)		0	11	13	3	
Overtime (Employees)		0	0	0	0	
Hires		9	0	8	0	17
Layoffs		0	0	0	13	13
Regular Cost ($)	800	28,000	19,200	34,400	21,600	103,200
Undertime Cost ($)	8	-	88	104	24	216
Overtime Cost ($)	15	-	-	-	-	-
Hiring Cost ($)	900	8,100	-	7,200	-	15,300
Layoff Cost ($)	1,200	-	-	-	15,600	15,600
Total Cost ($)		36,100	19,288	41,704	37,224	134,316

Figure 30 Mixed Strategy

Strategy	Total Cost	
Chase	$123,100	
Level	**$120,660**	✓
Mixed	$134,316	

Figure 31 Compere

Task IX

Conclusion

This project presented a comprehensive manufacturing systems engineering analysis of the Chicken Burger production line at **Nabil Factory**, integrating theoretical principles with real industrial data and software-based modeling. By applying tools such as Value Stream Mapping (VSM), Theory of Constraints (TOC), demand forecasting, capacity planning, MRP, scheduling, and resource allocation, the study provided a structured evaluation of the current production system and proposed realistic improvement strategies.

The analysis confirmed that the production system is **robust and delivery-focused**, with sufficient installed capacity and strong execution discipline. The freezer was clearly identified as the system bottleneck, governing throughput, cycle time, and resource allocation decisions. Current performance is achieved through **inventory buffers, batch processing, and overtime**, which ensure delivery reliability and food safety but also result in long lead times, high inventory levels, and increased operating costs.

The proposed future-state improvements and advanced planning framework demonstrated that significant performance gains can be achieved **without capital investment or compromising food safety**. By shifting from buffer-based compensation to **constraint-aware, flow-oriented planning**, the system can reduce lead time (15–25%), inventory (10–15%), and overtime (20–25%) while maintaining service levels. The selection of a **Level workforce strategy**, bottleneck-driven scheduling, rolling MRP, and strategic buffer rebalancing proved to be the most suitable and cost-effective approach for this high-volume, batch-based food manufacturing environment.

Overall, the project shows that Nabil Factory's production planning system is **effective but not fully optimized**. With disciplined integration of planning decisions across forecasting, capacity, inventory, scheduling, and resource allocation, the factory can evolve toward a more efficient, resilient, and sustainable production system

Limitations of the Current and Proposed Approach

Although the analysis and proposed planning framework provide clear improvements for the Chicken Burger production line, several limitations remain from a **company perspective**.

First, the planning system at Nabil Factory is still **highly dependent on inventory and overtime** as primary buffering mechanisms. While the proposed improvements reduce this dependence, they do not eliminate it entirely. This is largely due to external constraints such as long supplier lead times and strict food safety requirements, which limit flexibility in batch size and process sequencing.

Second, the production planning and resource allocation improvements are **line-specific**. Nabil Factory operates multiple production lines with a high product mix, and interactions between lines were not fully considered. In reality, shared resources such as labor, cold storage, and utilities may create additional constraints that are not captured when focusing on a single line.

Third, the planning system relies on **periodic decision-making** (weekly and monthly reviews). While this provides stability, it limits the factory's ability to respond quickly to unexpected disruptions such as supplier delays, equipment failures, or sudden market shocks. The current ERP-based planning approach supports execution well but lacks real-time decision intelligence.

Finally, while software tools such as Odoo and Microsoft Project provide strong structural support, **planning decisions still depend heavily on human judgment**. Variability in decision-making practices between planners and supervisors may reduce consistency and lead to suboptimal resource allocation under pressure.

Further Work and Improvement Opportunities for Nabil Factory

Several improvement opportunities remain that could further enhance production planning effectiveness at Nabil Factory One important area for future development is **supplier integration and lead-time reduction**.Closer collaboration with regional suppliers or dual-sourcing strategies could significantly reduce raw material lead times, allowing lower safety stock levels and shorter overall lead times without increasing risk.

Another opportunity lies in **advanced scheduling and simulation**. Introducing discrete-event simulation or advanced planning and scheduling tools would allow the factory to test different scheduling rules, batch sizes, and buffer policies under realistic variability before implementing changes on the shop floor and **real-time performance monitoring** could strengthen decision make Integrating KPIs such as bottleneck utilization WIP levels overtime usage and delivery performance into live dashboards would allow managers to detect deviations early and adjust plans proactively rather than reactively.

From a resource perspective, further work could focus on **multi-skilling and workforce flexibility**. Expanding operator cross-training would improve labor allocation flexibility around the bottleneck and reduce reliance on overtime during peak demand periods.

Finally, future efforts should consider **sustainability and energy efficiency**, particularly in freezing and cold storage operations. Optimizing energy consumption while maintaining food safety could reduce operating costs and support the company’s long-term environmental objectives.